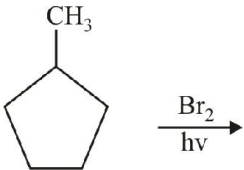
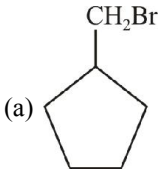
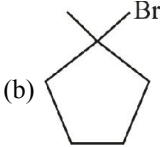
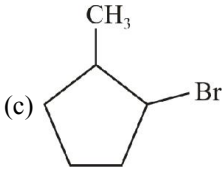
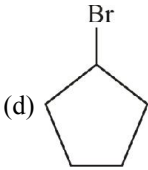
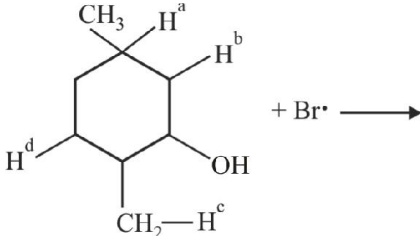
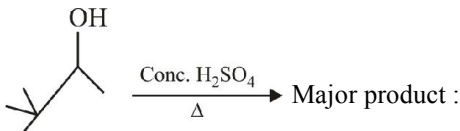
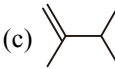


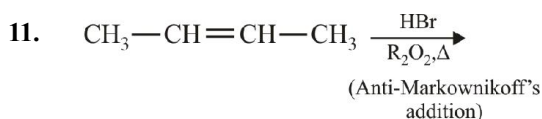
HYDROCARBONS

Table of Contents

↗	Exercise - 1 : Basic Objective Questions	2
↗	Exercise - 2 : Previous Year JEE Mains Questions	15
↗	Exercise - 3 : Advanced Objective Questions	19
↗	Exercise - 4 : Previous Year JEE Advanced Questions	37
↗	Answer Key	43

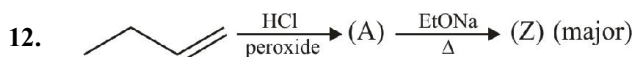
EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

1. The decreasing order of the anti-knocking value of octane number of the following is :
(I) CH_4 (II) C_2H_6 (III) C_3H_8 (IV) C_4H_{10}
(a) (I) > (II) > (III) > (IV) (b) (IV) > (III) > (II) > (I)
(c) (I) > (III) > (II) > (IV) (d) (IV) > (II) > (III) > (I)
2. On catalytic reduction (H_2/Pt) how many alkenes will give 2-methylbutane ?
(a) 1 (b) 2
(c) 3 (d) 4
3. Which of the following compounds can be best prepared by Wurtz-reaction ?
(a) Iso-butane (b) n-butane
(c) n-pentane (d) Iso-pentane
4. Select the structure of the major product formed in the reaction
- 
- (a)  (b) 
(c)  (d) 
5. (A) + $\text{Cl}_2 \xrightarrow{h\nu}$ monochloro product
To maximise the yield of monochloro product in the above reaction ?
(a) Cl_2 must be added in excess
(b) Reactant (A) must be added in excess
(c) Reaction must be carried out in dark
(d) Reaction must be carried out with equimolar mixture of Cl_2 and A
6. Select the chain propagation steps in the free-radical chlorination of methane.
(1) $\text{Cl}_2 \rightarrow 2\text{Cl}^\bullet$
(2) $\text{Cl}^\bullet + \text{CH}_4 \rightarrow \text{CH}_3\text{Cl} + \text{H}^\bullet$
(3) $\text{Cl}^\bullet + \text{CH}_4 \rightarrow \text{CH}_3^\bullet + \text{HCl}$
(4) $\text{H}^\bullet + \text{Cl}_2 \rightarrow \text{HCl} + \text{Cl}^\bullet$
(5) $\text{CH}_3^\bullet + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{Cl}^\bullet$
(a) 2, 3, 5 (b) 1, 3, 6
(c) 3, 5 (d) 2, 3, 4
7. 
 $\text{Br}^\bullet$ will abstract which of the hydrogen most readily.
(a) a (b) b
(c) c (d) d
8. $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3 \xrightarrow[\Delta]{\text{AlCl}_3} \text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
Above reaction is an example of :
(a) isomerization (b) polymerization
(c) cracking (d) de-hydrogenation
9. Which of the following is the major product when 1-butanol is heated with concentrated H_2SO_4 ?
(a) 1-butene (b) Cis-2-butene
(c) Trans-2-butene (d) All of the above
10. 
Major product :
(a)  (b) 
(c)  (d) None of these

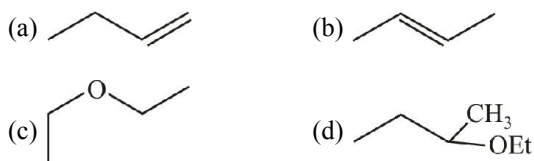


Comment on optical activity of the products :

- (a) Racemic (b) Diastereomer
(c) Meso (d) Optically pure enantiomer



Identify (Z) in the above sequence of reactions :

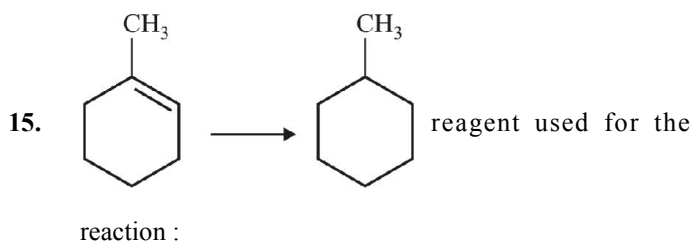


13. Which compound would be the major product when $(\text{CH}_3)_2\text{CHCHBrCH}_2\text{Br}$ is treated with alcoholic KOH ?

- (a) $(\text{CH}_3)_2\text{CHCH}_2\text{CHOHCH}_3$
(b) $(\text{CH}_3)_2\text{CHCH}(\text{OH})\text{CH}_2\text{CH}_2\text{OH}$
(c) $(\text{CH}_3)_2\text{COCH}_2\text{CH}_2\text{CH}_3$
(d) $(\text{CH}_3)_2\text{C}=\text{CHCH}_2\text{OH}$

14. On catalytic reduction with H_2/Pt how many alkenes will give n-butane ?

- (a) 1 (b) 2
(c) 3 (d) 4



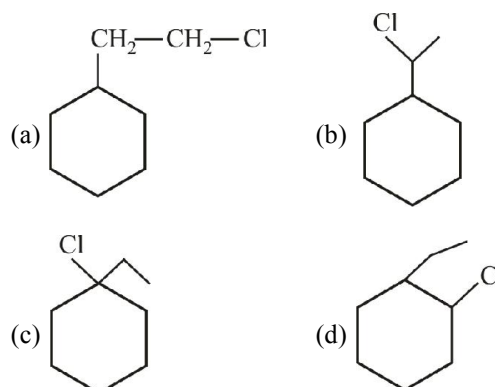
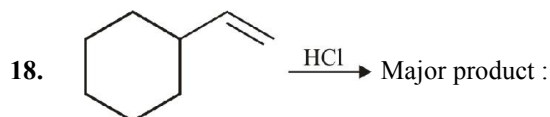
- (a) LiAlH_4 (b) H_2/Ni
(c) NaBH_4 (d) $\text{H}_2, \text{Pd} - \text{BaSO}_4$

16. Give the major product of the following reaction
 $3,3\text{-dimethyl-1-butene} + \text{HCl} \rightarrow \text{Major}$

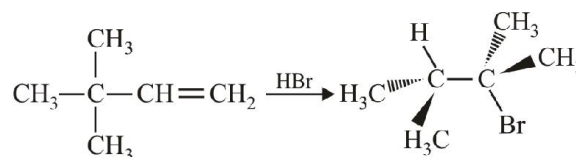
- (a) 2-chloro-3,3-dimethyl butane
(b) 2-chloro-2,3-dimethyl butane
(c) 1,2-dichloro-3,3-dimethylbutane
(d) 1-chloro-3,3-dimethylbutane

17. $\text{H}_3\text{C}-\text{CH}=\text{CH}_2 + \text{HCl} \xrightarrow{\text{Peroxide}}$ Product, the intermediate of reaction is :

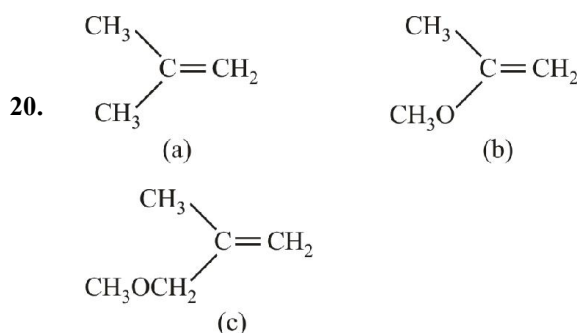
- (a) $\text{CH}_3-\dot{\text{C}}\text{H}-\text{CH}_3$ (b) $\text{CH}_3-\text{CH}_2-\dot{\text{C}}\text{H}_2$
(c) $\text{CH}_3-\text{CH}^+-\text{CH}_3$ (d) $\text{CH}_3-\text{CH}_2-\text{CH}_2^+$



19. Consider the addition of HBr to 3, 3-Dimethyl-1-butene shown below. What is the best mechanistic explanation for the formation of the observed product ?

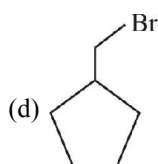
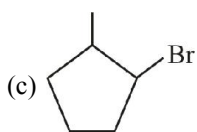
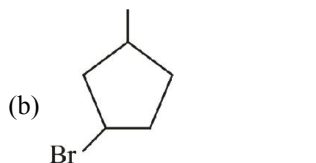
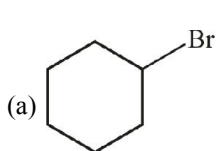
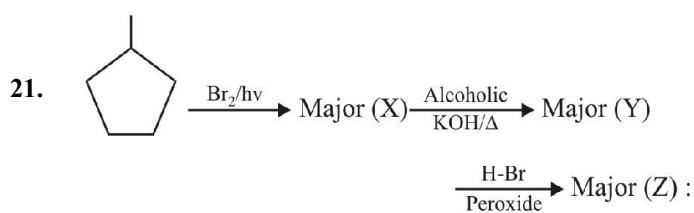


- (a) Protonation of the alkene followed by a hydride shift and addition of bromide to the carbocation
(b) Double bond shift in the alkene following by the protonation and addition of bromide to the carbocation
(c) Addition of bromide to the alkene followed by a double bond shift and protonation
(d) Protonation of the alkene followed by a methyl shift and addition of bromide to the carbocation

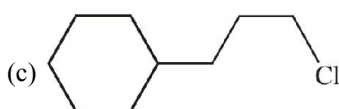
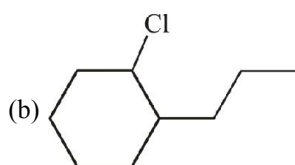
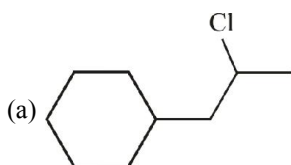
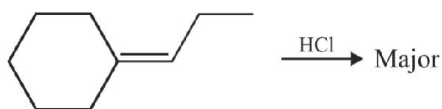


Arrange the above in the decreasing order of reactivity towards HBr :

- (a) $a > b > c$ (b) $b > a > c$
(c) $b > c > a$ (d) $a > c > b$



22. Choose the correct major product for the reaction below



23. $\text{Alkene} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{Alcohol}$

Which of the following statements about this reaction is correct ?

- (a) It is Markovnikov addition of water to alkene.
- (b) It involves carbonium ion rearrangement, if possible.
- (c) both of these
- (d) none of these

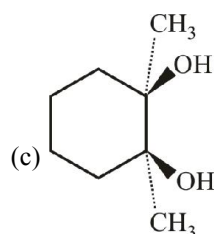
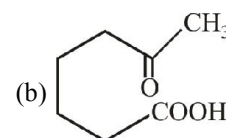
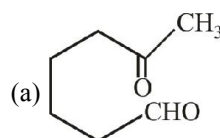
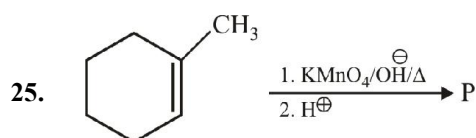
24. In case of addition of HX to an alkene, Markovnikov originally stated that

- (a) the H atom adds to the carbon atom of the double bond that already has greater number of H atoms and halide ion adds to the carbon atom of the double bond that has fewer number of H atoms.

(b) the H atom adds to the carbon atom of the double bond that has fewer number of H atoms and halide ion adds to the carbon atom of the double bond that has greater number of H atom.

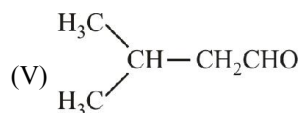
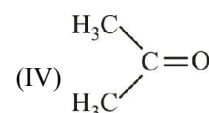
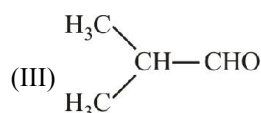
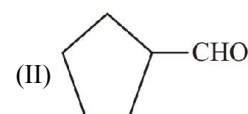
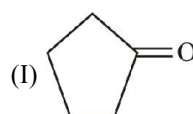
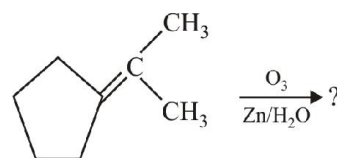
(c) both of these depending upon situation

(d) none of these



(d) none of these

26. Identify the product of the reaction when molecule undergoes ozonolysis



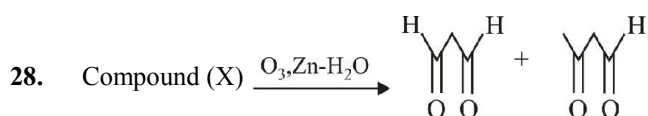
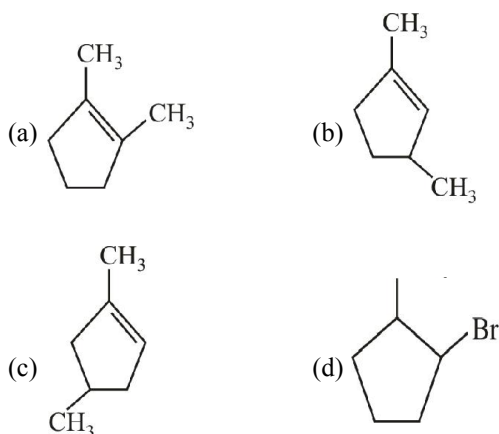
(a) I is only product

(b) I and III

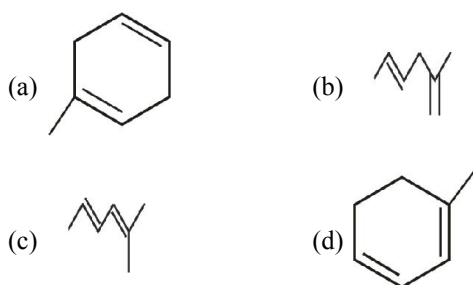
(c) I and IV

(d) II and IV

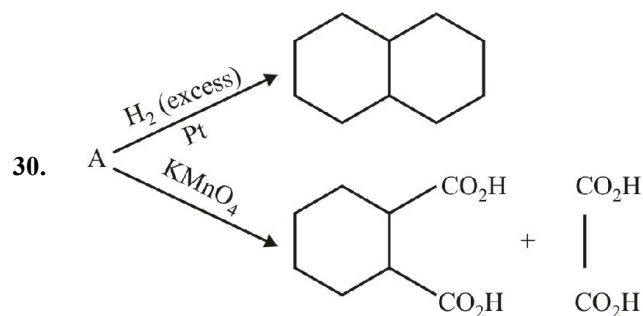
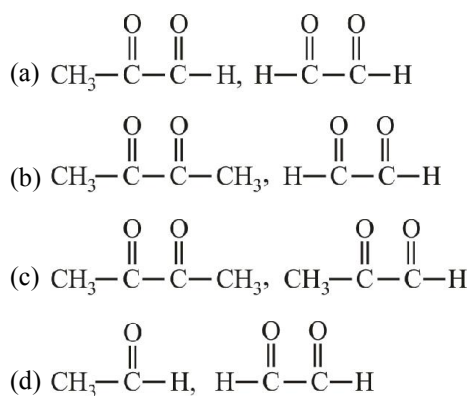
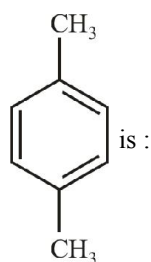
27. Which compound will yield 5-keto-2-methyl hexanal upon treatment with O_3 ?



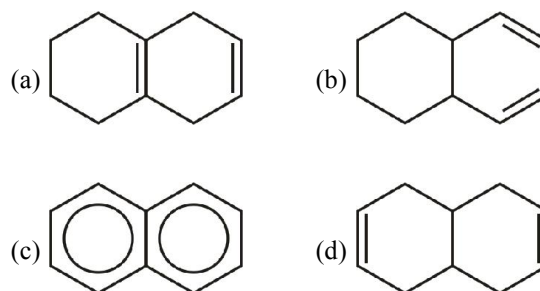
Find the structure of (X).



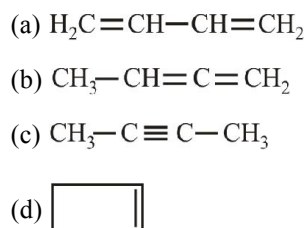
29. Ozonolysis of the following compound will produce;



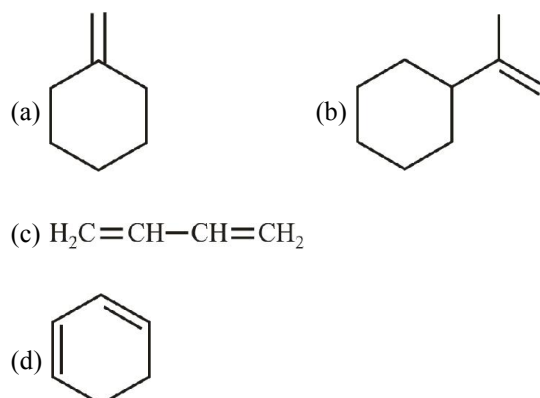
Compound (A) is :

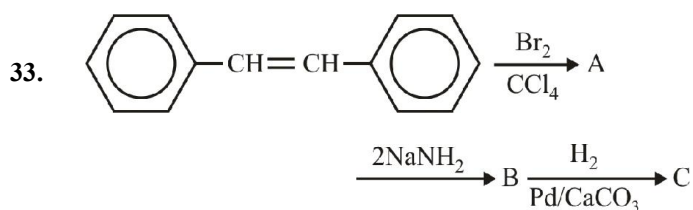


31. An organic compound C_4H_6 on ozonolysis give $HCHO$, CO_2 , CH_3CHO . Compound will be :

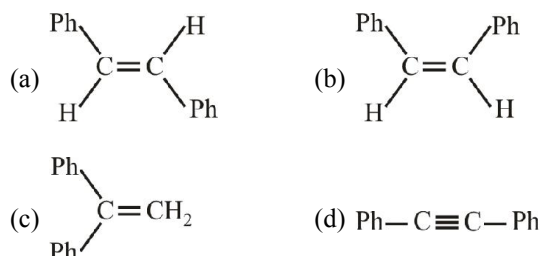


32. Which of the following compound does not evolve CO_2 gas, when undergo oxidative ozonolysis ?

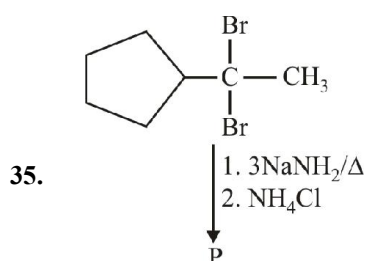
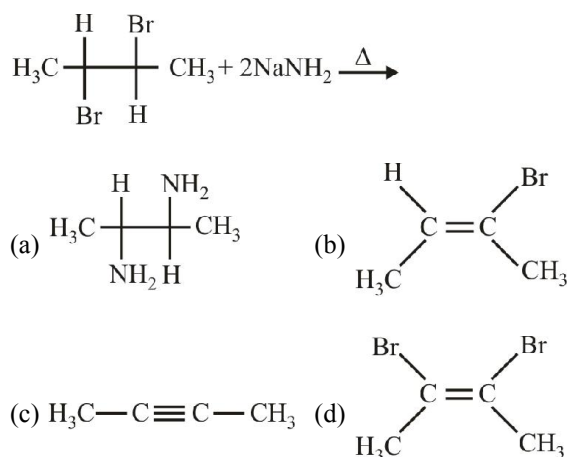




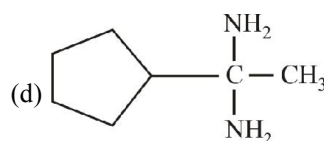
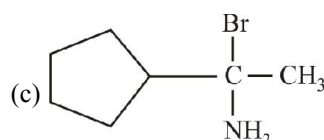
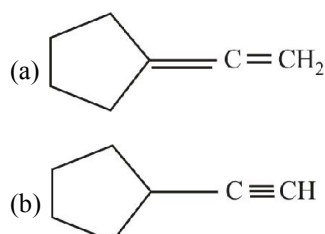
Product (C) is



34. Identify the major product



P should be



36. Rank the following compounds with respect to increasing acidity

- (1) 1-butene (2) butane
(3) 1-butyne (4) NH_3
(a) $4 < 3 < 1 < 2$ (b) $2 < 1 < 4 < 3$
(c) $2 < 1 < 3 < 4$ (d) $3 < 1 < 2 < 4$

37. Which of the following reacts with Tollen's reagent giving white precipitate ?

- (a) 2-pentene (b) 2-pentyne
(c) 1-butyne (d) 1-butanol

38. Which of the following reagents can be distinguish propyne from propene ?

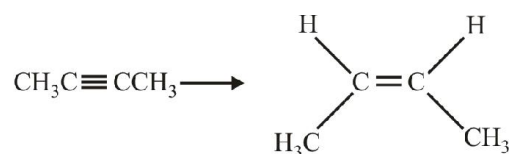
- (a) Br_2 , CCl_4 (b) Dilute KMnO_4
(c) Conc. H_2SO_4 (d) AgNO_3 in NH_4OH

39. Which is the most suitable reagent among the following distinguish compound (3) from the others ?

- (1) $\text{CH}_3\text{C}\equiv\text{C}-\text{CH}_3$ (2) $\text{CH}_3\text{CH}_2-\text{CH}_2-\text{CH}_3$
(3) $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$ (4) $\text{CH}_3\text{CH}=\text{CH}_2$

- (a) Bromine in carbon tetrachloride
(b) Bromine in acetic acid solution
(c) Alk. KMnO_4
(d) Ammonical silver nitrate

40. Choose the reagent to carry out the reaction.

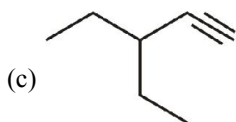
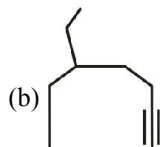
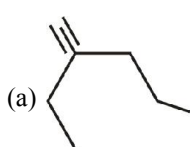


- (a) H_2 + Lindlar catalyst (b) $\text{Li} + \text{NH}_3$
(c) Conc. H_2SO_4 (d) $\text{H}_2\text{O} + \text{H}^+$

41. Which of the following reacts with Tollen's reagent giving white precipitate ?

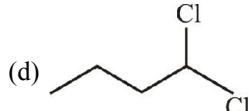
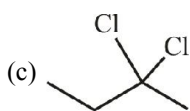
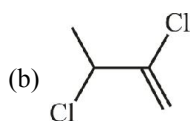
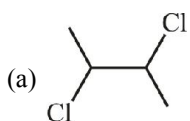
- (a) 2-pentene (b) 2-pentyne
(c) 1-butyne (d) 1-butanol

42. Which alkyne gives 3-ethylhexane on catalytic hydrogenation ?



(d) All of these

43. Predict the product of the reaction 1-butyne with 2 equivalents of HCl

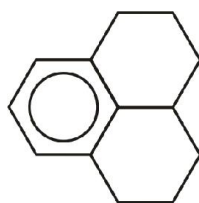


44. $\text{CaC}_2 \xrightarrow{\text{H}_2\text{O}} (\text{A}) \xrightarrow{\text{Red hot Cu tube}} (\text{B}),$
(Calcium carbide)

Product (B) of the reaction is :

- (a) Toluene (b) Ethyl-benzene
(c) Benzene (d) Butyne

45. How many benzylic hydrogens are present in the hydrocarbon shown ?



- (a) 3 (b) 5
(c) 4 (d) 6

46. The 'middle oil' fraction of coal-tar distillation contains

- (a) benzene (b) naphthalene
(c) anthracene (d) xylene

47. On heating a mixture of sodium benzoate and sodalime, the following is obtained

- (a) toluene (b) phenol
(c) benzene (d) benzoic acid

48. Catalytic dehydrogenation of n-heptane in the presence of $\text{Cr}_2\text{O}_3/\text{Al}_2\text{O}_3$ at 750 K gives

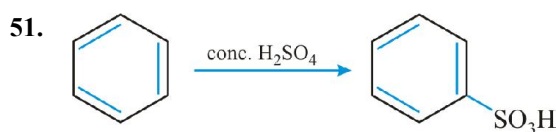
- (a) iso-heptane (b) 1-heptane
(c) toluene (d) 2, 3-dimethyl pentene-1

49. When sodium benzoate is heated with soda lime, the product formed is :

- (a) phenol (b) chlorobenzene
(c) benzene (d) benzaldehyde

50. Among the following compounds that can be most readily nitrated is :

- (a) benzene (b) benzaldehyde
(c) toluene (d) chlorobenzene



The electrophile in the reaction is

- (a) SO_3H^+ (b) SO_2
(c) SO_3 (d) H_2SO_4

52. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 . In the nitrating mixture, HNO_3 acts as a/an

- (a) base (b) acid
(c) reducing agent (d) catalyst

53. Which of the following compounds will not undergo Friedel-Craft acylation when treated with $\text{CH}_3\text{CH}_2\text{COCl}$, AlCl_3 ?

- (a) Toluene (b) p-Xylene
(c) Chlorobenzene (d) Benzophenone

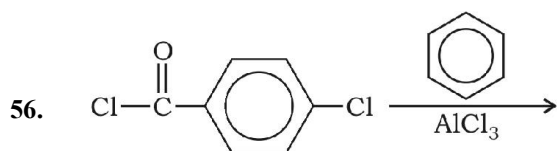
54. In the Gattermann-Koch synthesis, the electrophile is

- (a) $:\text{C}\equiv\text{O}^+$ (b) $\text{H}-\text{C}^+=\text{O}$

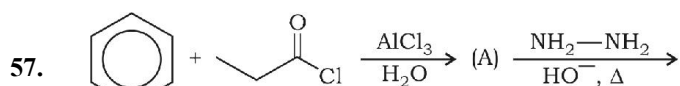
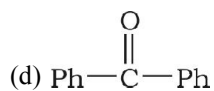
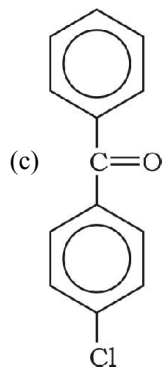
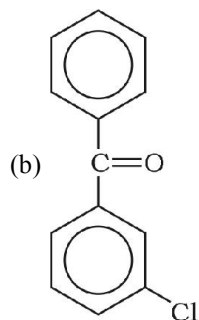
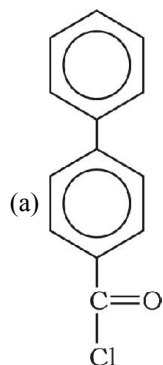
- (c) $\text{H}-\text{C}(=\text{O})-\text{Cl}$ (d) Cl^+

55. In desulfonation of benzene sulfonic acid, on treatment with steam, electrophile is

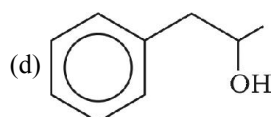
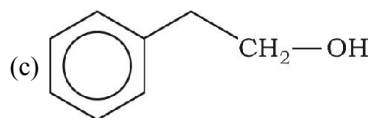
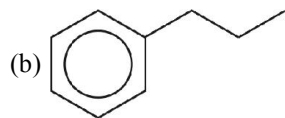
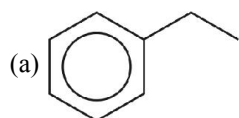
- (a) H^+ (b) SO_3
(c) H_2O (d) none of these



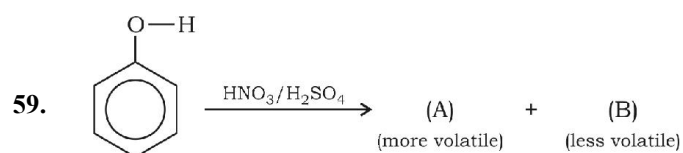
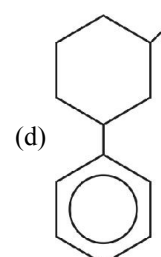
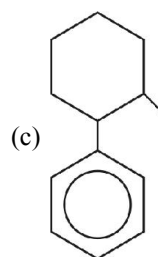
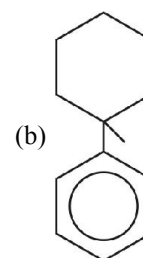
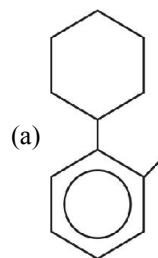
(A) ; Unknown (A) is



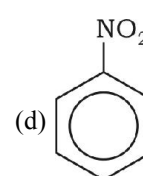
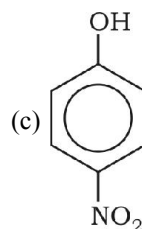
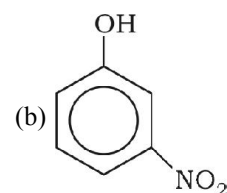
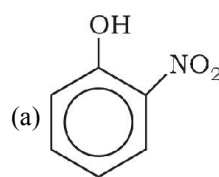
(B), Product (B) in this reaction is

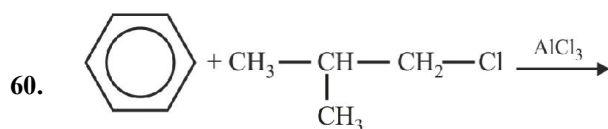


product of this reaction is

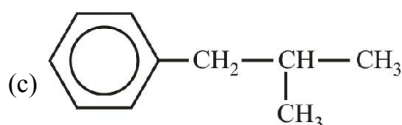
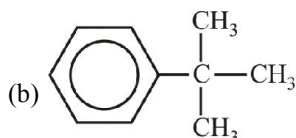
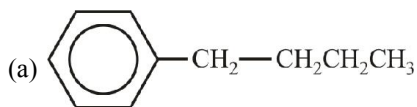


Product (A) of above reaction is



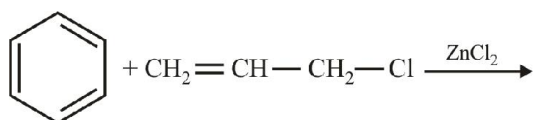


Compound X, is :

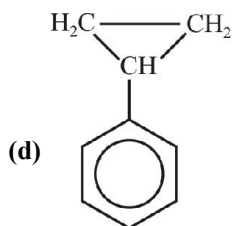
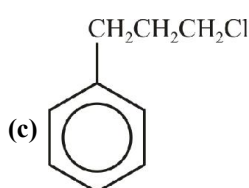
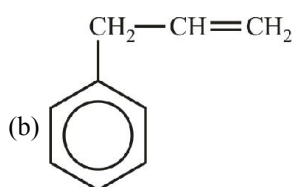
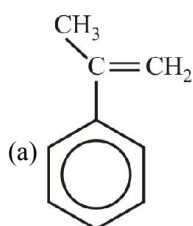


(d) all are correct

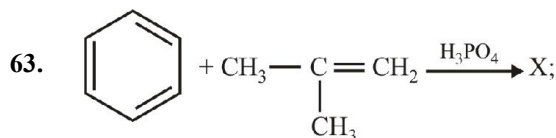
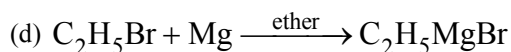
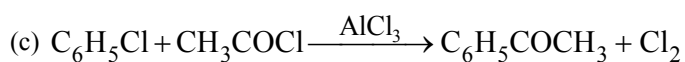
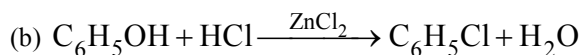
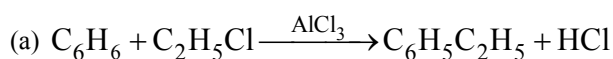
61. In the given reaction, what is the product ?



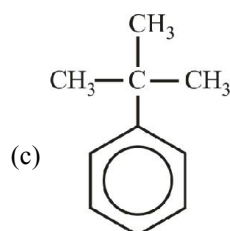
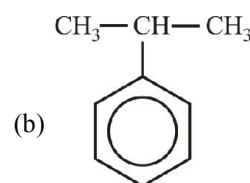
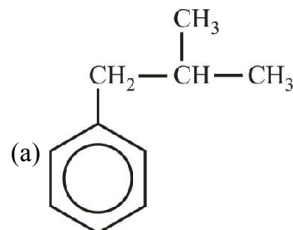
Product + HCl



62. Which of the following represents Friedel-Crafts reaction ?

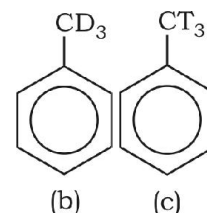
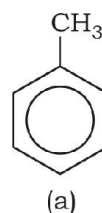


The compound X is :



(d) none of these

64. Arrange the following in decreasing order of reactivity towards EAS (electrophilic aromatic substitution)



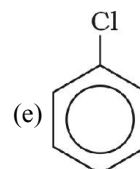
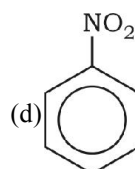
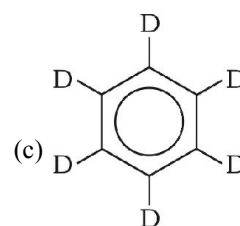
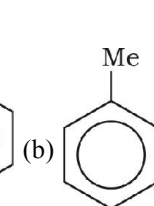
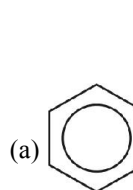
(a) $a > b > c$

(b) $c > b > a$

(c) $a > c > b$

(d) $c > a > b$

65. Arrange the following in decreasing order of rate of Nitration.



(a) $b < c < a < d < e$

(b) $d < e < a = c < b$

(c) $d < a = c < e < b$

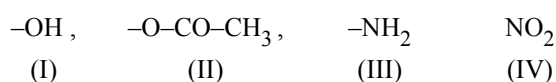
(d) $a < c < b < e < d$

66. Consider the following statements

- I : Hyperconjugation helps in increasing electron-density at ortho and para-sites in case of alkyl groups.
II : Electron-donating groups are activators and ortho and para-directors because they stabilise the arenium cations.
III : Electron-withdrawing groups are activators and are meta-directors because they stabilise the arenium cations.
IV : Halogens are activators and ortho and para-directors.
Select correct statements

- (a) I, II (b) I, II, III
(c) II, III, IV (d) I, IV

67. Arrange these groups in decreasing order of their electron-donating power in resonance



- (a) III > I > II > IV (b) II > I > III > IV
(c) III > II > I > IV (d) I > III > II > IV

68. In electrophilic aromatic substitution reaction, a phenyl substituent on the aromatic ring is

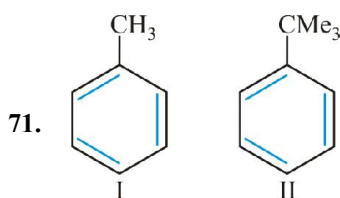
- (a) a deactivator and m-director
(b) a deactivator and o, p-director
(c) an activator and m-director
(d) an activator and o, p-director

69. In electrophilic aromatic substitution reactions, a $-\text{N}=\text{O}$ substituent on the aromatic ring is

- (a) a deactivator and m-director
(b) a deactivator and o, p-director
(c) an activator and m-director
(d) an activator and o, p-director

70. Which of the following is not the activating group ?

- (a) $-\text{NH}-\text{C}(=\text{O})-\text{R}$ (b) $-\text{O}-\text{C}(=\text{O})-\text{R}$
(c) $-\text{CH}=\text{CR}_2$ (d) $-\text{F}$



Which of the following is a correct comparison of rate of electrophilic substitution (bromination) ?

- (a) $r_I > r_{II}$ (b) $r_I < r_{II}$
(c) $r_I = r_{II}$ (d) None of these

72. Which xylene gives only one monochloro derivative by electrophilic aromatic substitution ?

- (a) p-xylene (b) m-xylene
(c) o-xylene (d) none of these

73. The correct decreasing order of relative reactivity with E^+ is

- (a) aniline > acetanilide > benzene > acetophenone
(b) aniline > acetanilide > acetophenone > benzene
(c) aniline > benzene > acetanilide > acetophenone
(d) benzene > acetophenone > acetanilide > aniline

74. Which of the following undergoes electrophilic substitution reaction (nitration), the fastest ?

- (a) Toluene (b) Benzene
(c) Benzoic acid
(d) Nitrobenzene

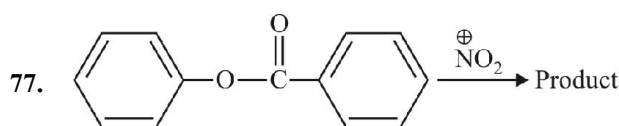
75. The correct decreasing order of relative reactivity with E^+ is

- (a) $p\text{-HOOC}-\text{C}_6\text{H}_4-\text{COOH} > p\text{-CH}_3\text{C}_6\text{H}_4\text{COOH} > \text{C}_6\text{H}_5\text{CH}_3 > p\text{-CH}_3\text{C}_6\text{H}_4\text{CH}_3$
(b) $p\text{-CH}_3\text{C}_6\text{H}_4\text{CH}_3 > \text{C}_6\text{H}_5\text{CH}_3 > p\text{-CH}_3\text{C}_6\text{H}_4\text{COOH} > p\text{-HOOC}-\text{C}_6\text{H}_4\text{COOH}$
(c) $\text{C}_6\text{H}_5\text{CH}_3 > p\text{-CH}_3\text{C}_6\text{H}_4\text{CH}_3 > p\text{-CH}_3\text{C}_6\text{H}_4\text{COOH} > p\text{-HOOC}-\text{C}_6\text{H}_4\text{COOH}$
(d) $p\text{-CH}_3\text{C}_6\text{H}_4\text{COOH} > \text{C}_6\text{H}_5\text{CH}_3 > p\text{-CH}_3\text{C}_6\text{H}_4\text{CH}_3 > p\text{-HOOC}-\text{C}_6\text{H}_4\text{COOH}$

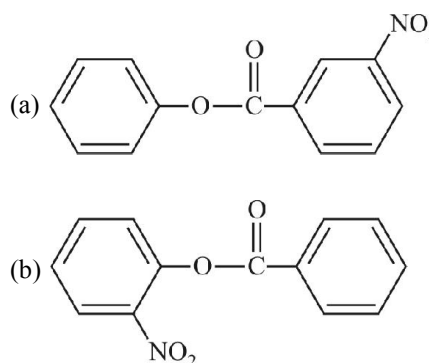
76. Benzene $\longrightarrow$ ortho-chloronitrobenzene

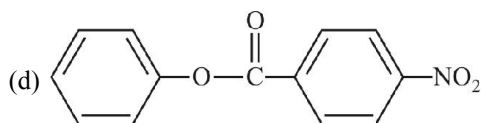
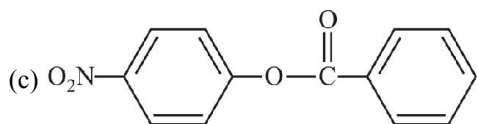
For the conversion, which of the following sequences is the most appropriate ?

- (a) Chlorination, then nitration
(b) Nitration then chlorination
(c) Chlorination, sulphonation, nitration, desulphonation
(d) Nitration, sulphonation, chlorination, desulphonation



Product of this reaction on mono-nitration is





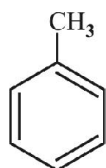
78. Which of the following compounds will undergo bromination most rapidly using Br_2 , FeBr_3 ?

- (a) p-methylacetanilide (b) bromobenzene
(c) acetanilide
(d) benzenesulphonic acid

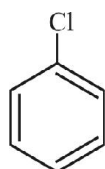
79. Identify the correct order of reactivity in electrophilic substitution reactions of the following compounds



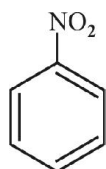
I



II



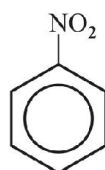
III



IV

- (a) $\text{I} > \text{II} > \text{III} > \text{IV}$ (b) $\text{IV} > \text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{I} > \text{III} > \text{IV}$ (d) $\text{II} > \text{III} > \text{I} > \text{IV}$

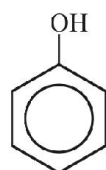
80. Arrange the following compounds in the order of increasing tendency to undergo electrophilic substitution.



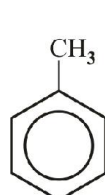
(1)



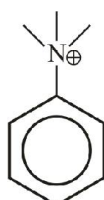
(2)



(3)



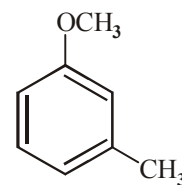
(4)



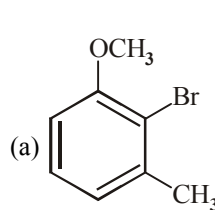
(5)

- (a) $5 < 1 < 2 < 4 < 3$ (b) $3 > 2 > 1 > 4 > 5$
(c) $1 > 2 > 5 > 3 > 4$ (d) $5 < 1 < 4 < 2 < 3$

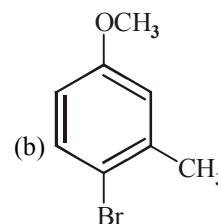
81. The major product obtained on the monobromination (with $\text{Br}_2/\text{FeBr}_3$) of the following compound A is



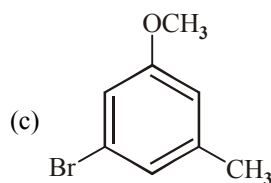
(A)



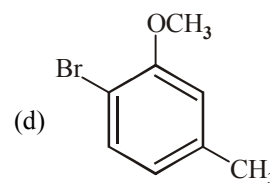
(a)



(b)



(c)

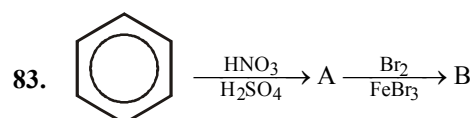


(d)

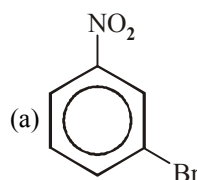
82. The order of decreasing reactivity towards an electrophilic reagent, for the following

- (1) Benzene (2) Toluene
(3) Chlorobenzene (4) Phenol
would be

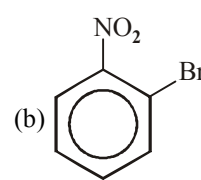
- (a) $1 > 2 > 3 > 4$ (b) $2 > 4 > 1 > 3$
(c) $4 > 3 > 2 > 1$ (d) $4 > 2 > 1 > 3$



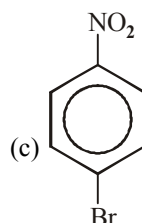
The compound B is



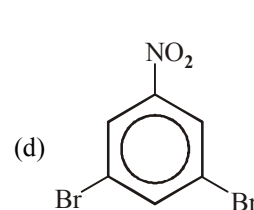
(a)



(b)

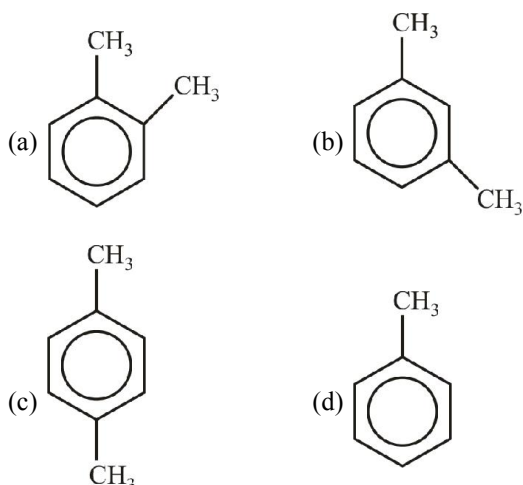


(c)

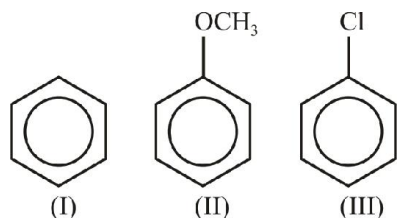


(d)

84. Which one of the following compounds give only one isomer upon nitration ?



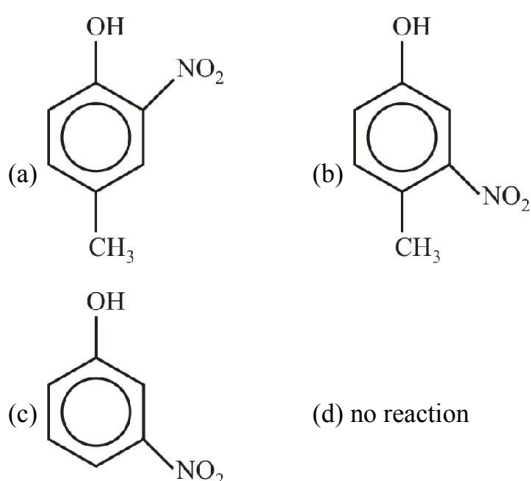
85. Consider the following compound



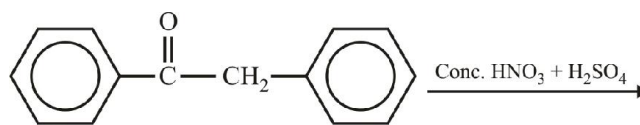
The relative reactivity towards halonium ion is such that :

- (a) I > II > III (b) II > I > III
(c) II > III > I (d) III > I > II

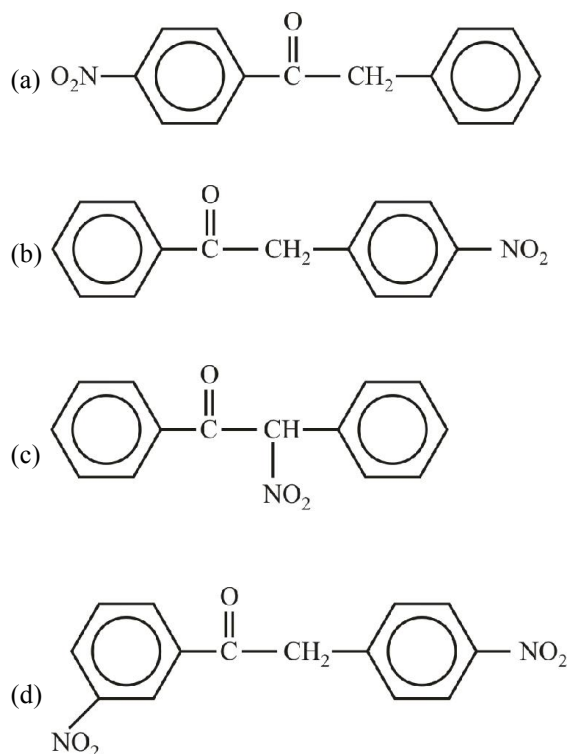
86. $\xrightarrow{\oplus \text{NO}_2}$ A; A can be :



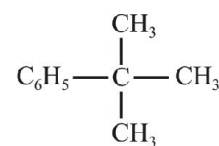
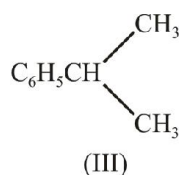
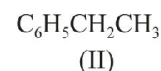
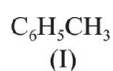
87. The major product obtained in the reaction



is expected to be :



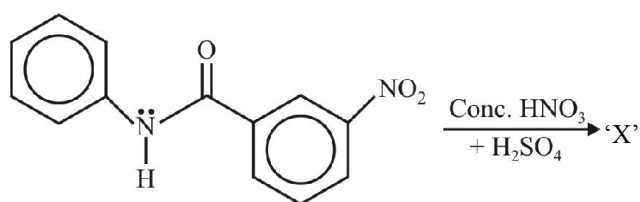
88. The order of reactivity of the following compounds



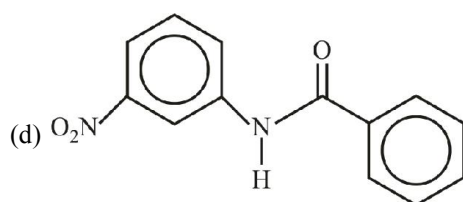
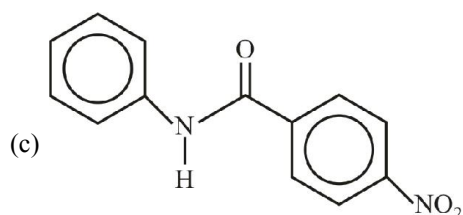
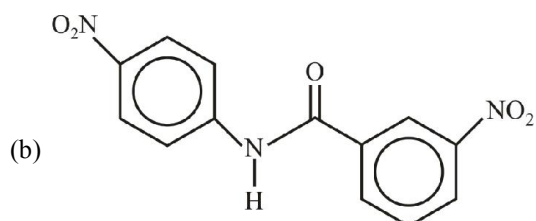
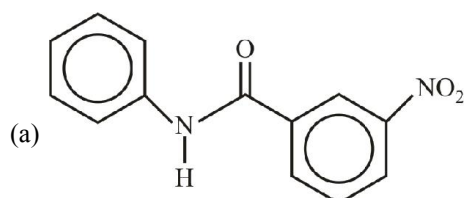
towards electrophilic substitution will be :

- (a) I > II > III > IV (b) IV > III > II > I
(c) II > I > III > IV (d) III > II > I > IV

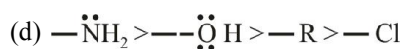
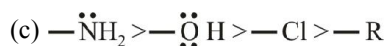
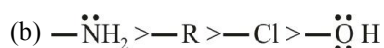
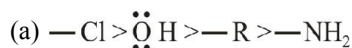
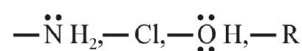
89. In the following reaction



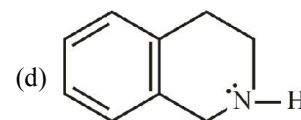
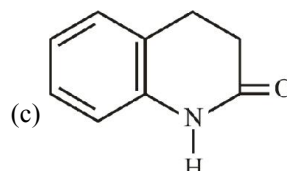
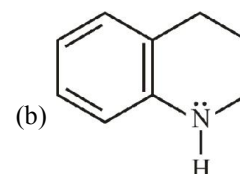
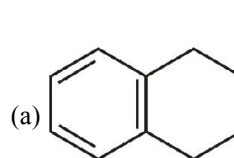
the structure of major product 'X' is :



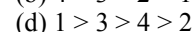
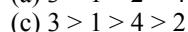
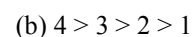
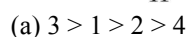
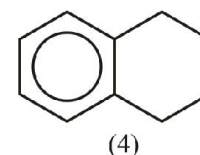
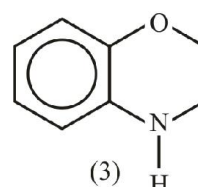
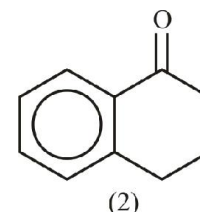
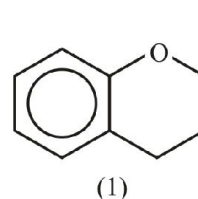
90. Arrange the following groups in order of decreasing o- and p-directing strength :



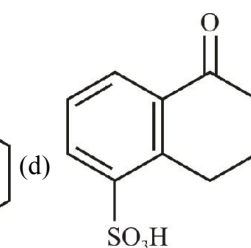
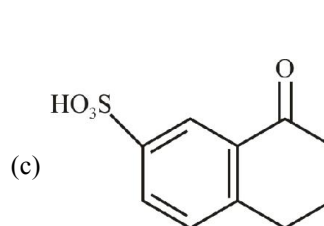
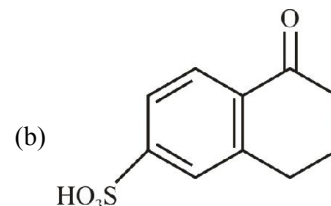
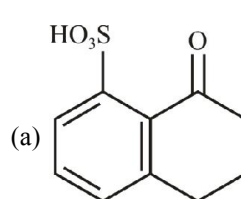
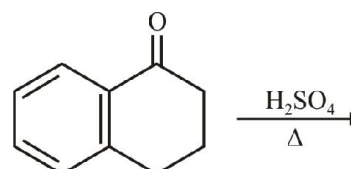
91. Which of the following compounds undergoes nitration at fastest rate ?



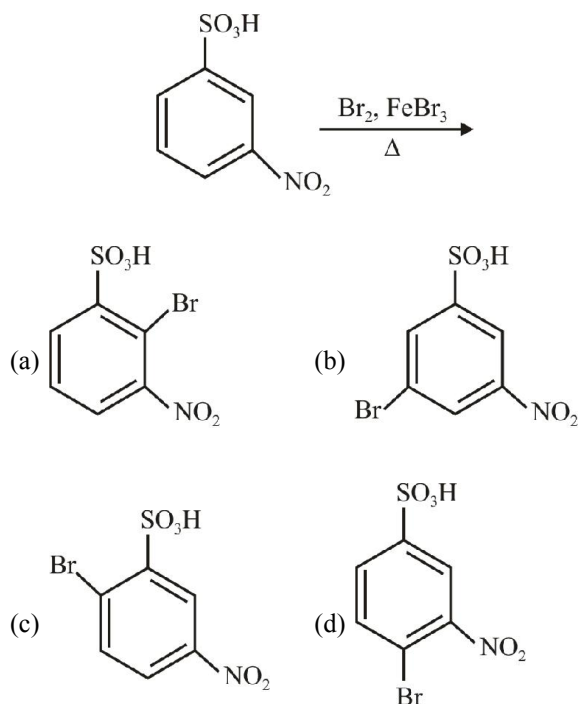
92. Rank the following compounds in decreasing order of reactivity in electrophilic aromatic substitution reaction :



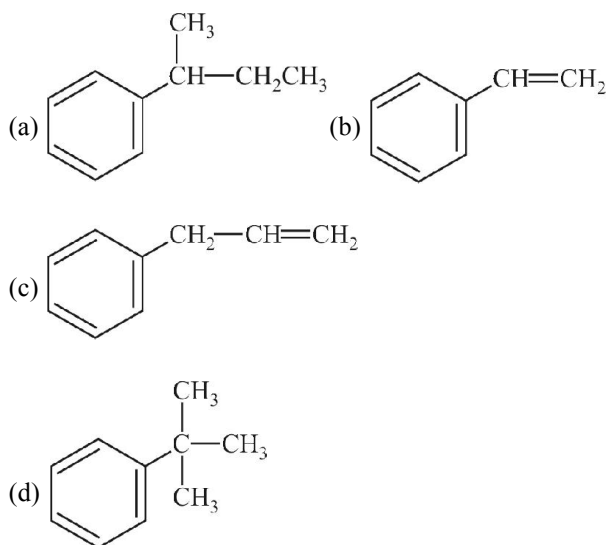
93. Which of the following is the major product from the sulphonation of α -tetralone ?



94. Which of the following is the major product from bromination of meta-nitrobenzene sulphonic acid ?

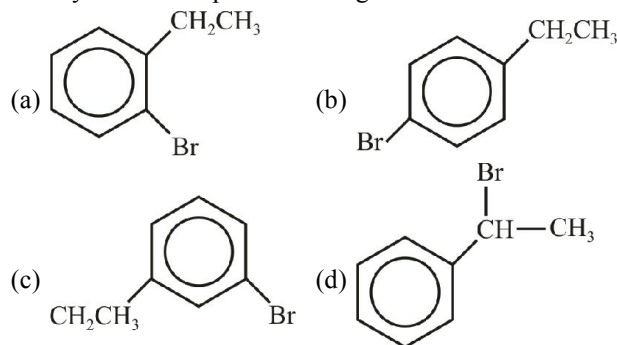


95. Gammexane can be obtained from benzene with reagent
- (a) Br_2 , sunlight (b) Cl_2 , sunlight
- (c) Cl_2 , CCl_4 (d) COCl_2 , anhydrous AlCl_3
96. Which of the following cannot be oxidised to benzoic acid on treatment with KMnO_4 , H^+ ?

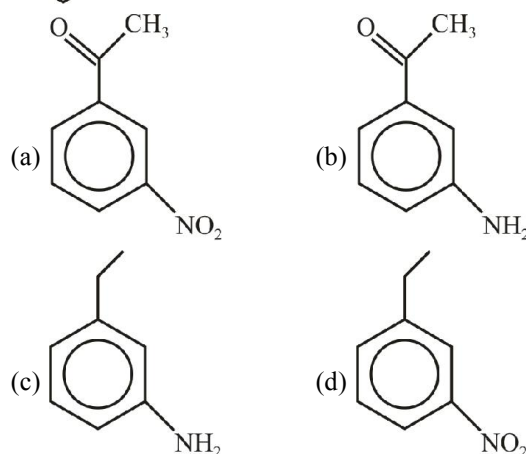


97. The product of following reaction is $\text{C}_6\text{H}_6 + \text{Cl}_2 \xrightarrow{h\nu}$?
- (a) $\text{C}_6\text{H}_5\text{Cl}$ (b) ortho $\text{C}_6\text{H}_4\text{Cl}_2$
- (c) $\text{C}_6\text{H}_6\text{Cl}_6$ (d) para $\text{C}_6\text{H}_4\text{Cl}_2$

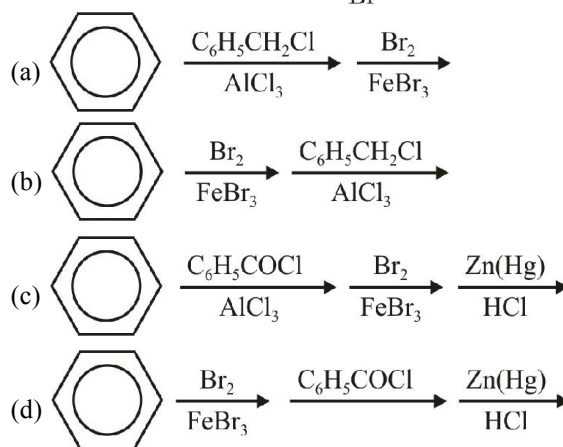
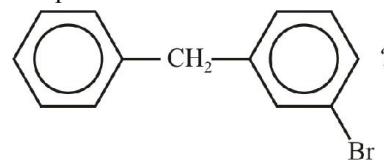
98. Which of the following will be obtained by the bromination of ethylbenzene in presence of light ?



99. CC(=O)c1ccccc1>HNO3, H2SO4>X>Zn(Hg), HCl>Y; Identify 'Y' :



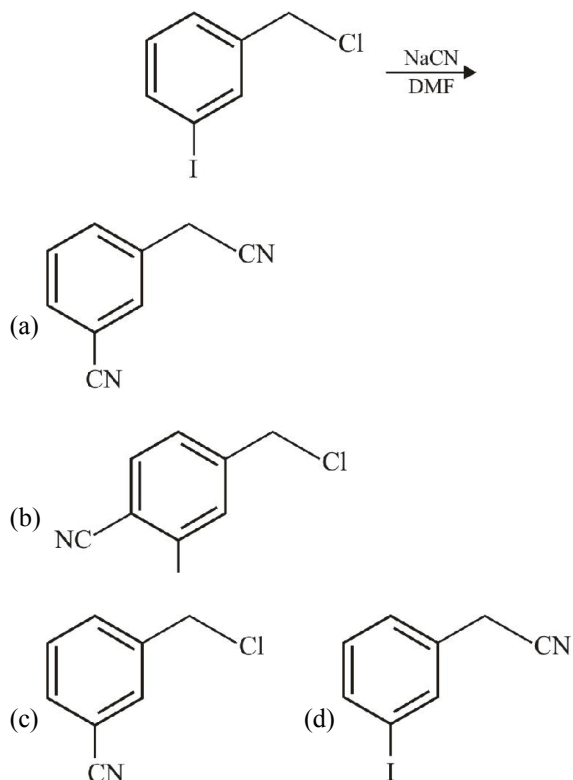
100. Which sequence of steps describes the best synthesis of compound

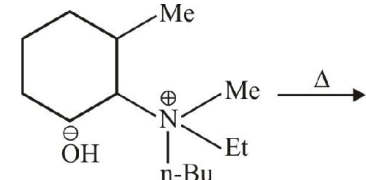
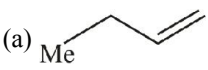
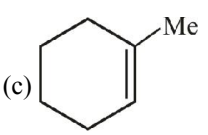
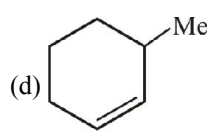


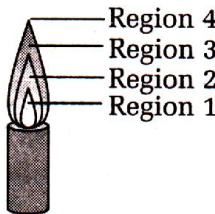
EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

- CH₃MgI is an organometallic compound due to (2002)
 - Mg—I bond
 - C—I bond
 - C—Mg bond
 - C—H bond
- Acetylene reacts with hypochlorous acid to form (2002)
 - Cl₂CHCHO
 - ClCH₂COOH
 - CH₃COCI
 - ClCH₂CHO
- Acetylene does not react with (2002)
 - Na
 - ammoniacal AgNO₃
 - HCl
 - NaOH
- On vigorous oxidation by permanganate solution (CH₃)₂C=CHCH₂CHO gives (2002)
 - (CH₃)₂CO and OHCCH₂CHO
 - (CH₃)₂CO and $\begin{array}{c} \text{C} \quad \text{CH} \quad \text{CH}_2\text{CHO} \\ | \quad | \\ \text{OH} \quad \text{OH} \end{array}$
 - (CH₃)₂CO and OHCCH₂COOH
 - (CH₃)₂CO and CH₂(COOH)₂
- The reaction of chloroform with alcoholic KOH and p-toluidine form (2003)
 -
 -
 -
 -
- Bottles containing C₆H₅I and C₆H₅CH₂I lost their original labels. They were labelled A and B for testing. A and B were separately taken in test tubes and boiled with NaOH solution. The end solution in each tube was made acidic with dilute HNO₃ and then some AgNO₃ solution was added. Substance B gave a yellow precipitate. Which one of the following statements is true for this experiment ? (2003)
 - A was C₆H₅I
 - A was C₆H₅CH₂I
 - B was C₆H₅I
 - Addition of HNO₃ was unnecessary
- Butene-1 may be converted to butane by reaction with (2003)
 - Zn-HCl
 - Sn-HCl
 - Zn-Hg
 - Pd/H₂
- On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be (2003)
 - propane
 - pentane
 - isopentane
 - neopentane
- On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be (2003)
 - gammexane
 - DDT
 - freon
 - hexachloroethane
- Acetyl bromide reacts with excess of CH₃MgI followed by treatment with a saturated solution of NH₄Cl gives (2004)
 - acetone
 - acetamide
 - 2-methyl-2-propanol
 - acetyl iodide
- On mixing ethyl acetate with aqueous sodium chloride, the composition of the resultant solution is (2004)
 - CH₃COOC₂H₅ + NaCl
 - CH₃COONa + C₂H₅OH
 - CH₃COCl + C₂H₅OH + NaOH
 - CH₃Cl + C₂H₅COONa
- Which one of the following has the minimum boiling point ?
 - n-butane
 - 1-butyne
 - 1-butene
 - Isobutene
- Which one of the following is reduced with zinc and hydrochloric acid to give the corresponding hydrocarbon ?
 - Ethyl acetate
 - Acetic acid
 - Acetamide
 - Butan-2-one
- Reaction of on molecule of HBr with one molecule of 1,3-butadiene at 40°C gives predominantly (2005)
 - 1-bromo-2-butene under kinetically controlled conditions
 - 3-bromobutene under thermodynamically controlled conditions
 - 1-bromo-2-butene under thermodynamically under kinetically controlled conditions
 - 3-bromobutene under kinetically controlled conditions

15. Elimination of bromine from 2-bromobutane results in the formation of (2005)
(a) predominantly 2-butyne
(b) predominantly 1-butene
(c) predominantly 2-butene
(d) equimolar mixture of 1 and 2-butene
16. Of the five isomeric hexanes, the isomer which can give two monochlorinated compounds, is (2005)
(a) 2-methylpentane
(b) 2, 2-dimethylbutane
(c) 2, 3-dimethylbutane
(d) n-hexane
17. 2-methyl butane on reacting with bromine in the presence of sunlight gives mainly (2005)
(a) 1-bromo-3-methylbutane
(b) 2-bromo-3-methylbutane
(c) 2-bromo-2methylbutane
(d) 1-bromo-2methylbutane
18. Alkyl halides react with dialkyl copper reagents to give (2005)
(a) alkenyl halies (b) alkanes
(c) alkyl copper halides (d) alkenes
19. The structure of the major product formed in the following reaction is



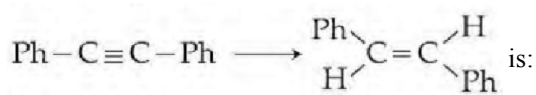
20. $\text{CH}_3\text{Br} + \text{Nu}^- \longrightarrow \text{CH}_3 - \text{Nu} + \text{Br}^-$
The decreasing order of the rate of the above reaction with nucleophile (Nu^-) A and D is :
[$\text{Nu} = (\text{A})\text{PhO}^-$, (B) AcO^- , (C) HO^- , (D) CH_3O^-]
(a) $\text{D} > \text{C} > \text{A} > \text{B}$ (b) $\text{D} > \text{C} > \text{B} > \text{A}$
(c) $\text{A} > \text{B} > \text{C} > \text{D}$ (d) $\text{B} > \text{D} > \text{C} > \text{A}$
21. 
- The alkene formed as a major product in the above elimination reaction is
(a)  (b) $\text{CH}_2=\text{CH}_2$
(c)  (d) 
22. Trans-2-phenyl-1-bromocyclopentane on reaction with alcoholic KOH produces (2006)
(a) 4-phenylcyclopentene (b) 2-phenylcyclopentene
(c) 1-phenylcyclopentene (d) 3-phenylcyclopentene
23. HBr reacts with $\text{CH}_2 = \text{CH} - \text{OCH}_3$ under anhydrous conditions at room temperature to give (2006)
(a) CH_3CHO and CH_3Br
(b) BrCH_2CHO and CH_3OH
(c) $\text{BrCH}_2 - \text{CH}_2 - \text{OCH}_3$
(d) $\text{H}_3\text{C} - \text{CHBr} - \text{OCH}_3$
24. The reaction of toluene with Cl_2 in presence of FeCl_3 gives predominantly (2007)
(a) benzoyl chloride (b) benzyl chloride
(c) o- and p-chlorotoluene
(d) m-chlorotoluene
25. Presence of a nitro group in a benzene ring (2007)
(a) activates the ring towards electrophilic substitution
(b) renders the ring basic
(c) deactivates the ring towards nucleophilic substitution
(d) deactivates the ring towards electrophilic substitution

26. Which of the following reactions will yield 2, 2-dibromopropane ?
- (a) $\text{CH}_3 - \text{C} \equiv \text{CH} + 2\text{HBr} \longrightarrow$
(b) $\text{CH}_3\text{CH} = \text{CHBr} + \text{HBr} \longrightarrow$
(c) $\text{CH} \equiv \text{CH} + 2\text{HBr} \longrightarrow$
(d) $\text{CH}_3 - \text{CH} = \text{CH}_2 + \text{HBr} \longrightarrow$
27. The treatment of CH_3MgX with $\text{CH}_3\text{C} \equiv \text{C} - \text{H}$ produces (2008)
- (a) $\text{CH}_3 - \text{CH} = \text{CH}_2$ (b) $\text{CH}_3\text{C} \equiv \text{C} - \text{CH}_3$
(c) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3$ (d) CH_4
28. The organic chloro compound, which shows complete stereochemical inversion during an $\text{S}_\text{N}2$ reaction is (2008)
- (a) $(\text{C}_2\text{H}_5)_2\text{CHCl}$ (b) $(\text{CH}_3)_3\text{CCl}$
(c) $(\text{CH}_3)_2\text{CHCl}$ (d) CH_3Cl
29. The hydrocarbon which can react with sodium in liquid ammonia is (2008)
- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_2\text{CH}_3$
(b) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CH}$
(c) $\text{CH}_3\text{CH} = \text{CHCH}_3$
(d) $\text{CH}_3\text{CH}_2\text{C} \equiv \text{CCH}_2\text{CH}_3$
30. In the following sequence of reactions, the alkene affords the compound 'B'
- $$\text{CH}_3\text{CH} = \text{CHCH}_3 \xrightarrow{\text{O}_3} \text{A} \xrightarrow[\text{Zn}]{\text{H}_2\text{O}} \text{B}$$
- The compound B is (2008)
- (a) $\text{CH}_3\text{CH}_2\text{CHO}$ (b) CH_3COCH_3
(c) $\text{CH}_3\text{CH}_2\text{COCH}_3$ (d) CH_3CHO
31. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is (2010)
- (a) propene (b) 1-butene
(c) 2-butene (d) ethene
32. Ozonolysis of an organic compound given formaldehyde as one of the products. This confirms the presence of (2011)
- (a) two ethylenic double bonds
(b) a vinyl group
(c) an iso-propyl group
(d) an acetylenic triple bond
33. 2-hexyne gives trans-2-hexene on treatment with (2012)
- (a) Pt/H_2 (b) Li/NH_3
(c) Pd/BaSO_4 (d) LiAlH_4
34. The major organic compound formed by the reaction of 1, 1, 1-trichloroethane with silver powder is (2014)
- (a) acetylene (b) ethene
(c) 2-butyne (d) 2-butene
35. The hottest region of Bunsen flame shown in the figure given below is (2016)
- 
- (a) region 2 (b) region 3
(c) region 4 (d) region 1
36. The trans-alkenes are formed by the reduction of alkynes with: (2018)
- (a) $\text{Sn} - \text{HCl}$ (b) $\text{H}_2 - \text{Pd/C}, \text{BaSO}_4$
(c) NaBH_4 (d) $\text{Na}/\text{liq. NH}_3$

JEE MAINS ONLINE QUESTION

1. The gas liberated by the electrolysis of Dipotassium succinate solution is: Online 2014 SET (2)
- (a) Ethane (b) Ethyne
(c) Propene (d) Ethene

2. The reagent needed for converting



Online 2014 SET (2)

- (a) LiAlH_4 (b) Cat. Hydrogenation
(c) $\text{H}_2/\text{Lindlar Cat.}$ (d) Li/NH_3
3. The major product obtained in the photo catalysed bromination of 2-methylbutane is:

Online 2014 SET (3)

- (a) 1-bromo-2-methylbutane
(b) 1-bromo-3-methylbutane
(c) 2-bromo-2-methylbutane
(d) 2-bromo-3-methylbutane
4. Which one of the following reaction is a method for the conversion of ketone into a hydrocarbon?

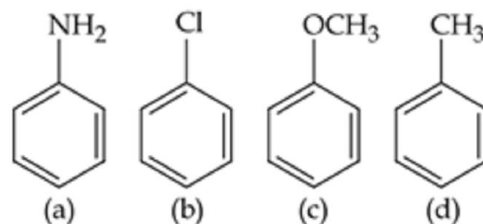
Online 2015 SET (1)

- (a) Aldol condensation
(b) Riemer-Tiemann reaction
(c) Cannizzaro reaction
(d) Wolf-Kishner reduction
5. The hydrocarbon with seven carbon atoms containing a neopentyl and a vinyl group is:

Online 2016 SET (1)

- (a) 2, 2-dimethyl-4-pentene
(b) Isopropyl-2-butene
(c) 4, 4-dimethylpentene
(d) 2, 2-dimethyl-3-pentene

6. The increasing order of nitration of the following compounds is :
Online 2018 SET (1)



- (a) (b) < (a) < (c) < (d)
(b) (a) < (b) < (c) < (d)
(c) (b) < (a) < (d) < (c)
(d) (a) < (b) < (d) < (c)

7. When 2-butyne is treated with $\text{H}_2/\text{Lindlar's catalyst}$, compound X is produced as the major product and when treated with Na/liq. NH_3 it produces Y as the major product. Which of the following statements is correct ?

(Online 2018 SET 2)

- (a) X will have higher dipole moment and higher boiling point than Y.
(b) Y will have higher dipole moment and higher boiling point than X.
(c) X will have lower dipole moment and lower boiling point than Y.
(d) Y will have higher dipole moment and lower boiling point than X.

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

- All questions marked "S" are single choice questions
- All questions marked "M" are multiple choice questions
- All questions marked "C" are comprehension based questions
- All questions marked "A" are assertion–reason type questions

(A) If both assertion and reason are correct and reason is the correct explanation of assertion.

(B) If both assertion and reason are true but reason is not the correct explanation of assertion.

(C) If assertion is true but reason is false.

(D) If reason is true but assertion is false.
- All questions marked "X" are matrix–match type questions
- All questions marked "I" are integer type questions

- (S) Which of the statements is correct ?

(I) Melting point of alkane increases with increase of C atoms and with increase in branching.

(II) Boiling point of alkane increases with increase of C atoms but with decrease in branching.

(III) Cycloalkanes have lower boiling point than normal alkane with same number of C atoms.

(IV) Alkenes have lower boiling point than same number of C atoms in alkanes.

(a) (I), (II) (b) (I), (II), (III)

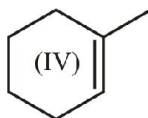
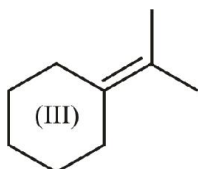
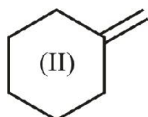
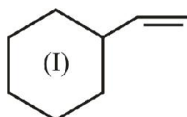
(c) (III), (IV) (d) (IV)

- (A) **Assertion :** Among isomeric alkanes, branching increase the boiling points.

Reason : Branching lowers the intermolecular van der Waal's force of attraction.

- (a) A (b) B
- (c) C (d) D

- (S) Arrange the following alkene in increasing order of their enthalpy of hydrogenation ($-\Delta H$)



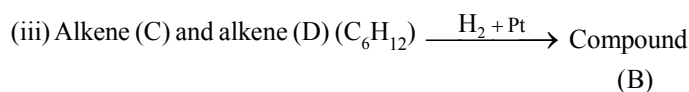
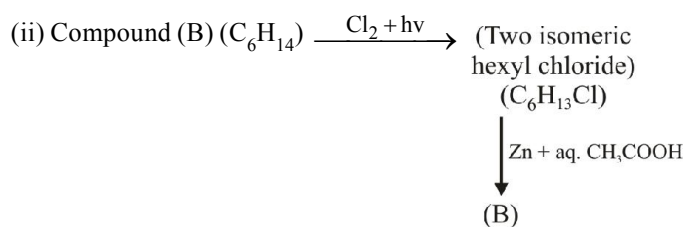
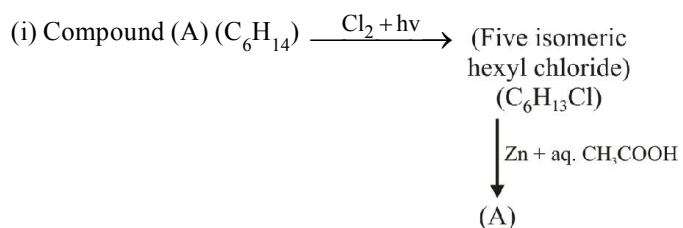
- (a) III < IV < II < I (b) III < IV < I < II
- (c) I < II < III < IV (d) I < II < IV < III

- (S) Which of the following would not form upon electrolysis of aqueous solution of potassium propanoate ?

- (a) Butane (b) Ethyl ethanoate
- (c) Ethyl propanoate (d) Ethene

- (I) How many monochlorinated products can be obtained from all the isomeric alkanes having the formula C_5H_{10} (excluding stereoisomers) ?

Passage



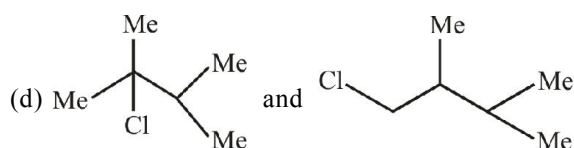
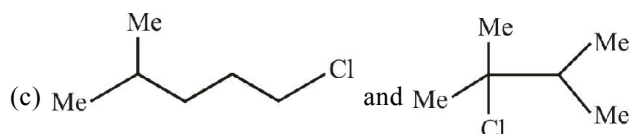
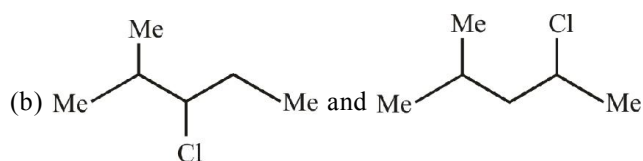
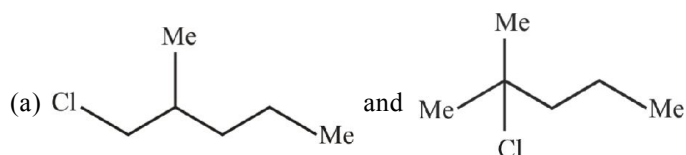
6. (C) Compound (A) is :

- (a) 2, 3-Dimethylbutane (b) 2-Methylpentane
(c) 2,2-Dimethylbutane (d) 3-Methylpentane

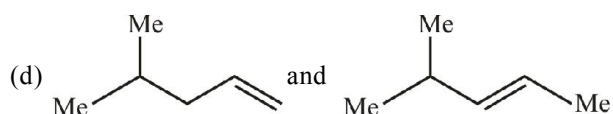
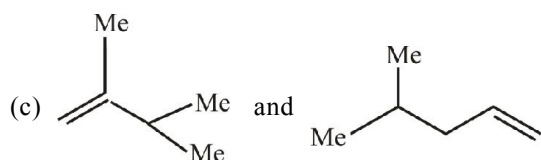
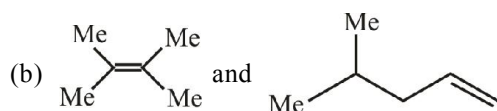
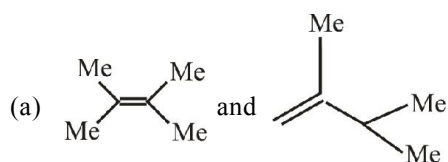
7. (C) Compound (B) is :

- (a) 2,3-Dimethylbutane (b) 2-Methylpentane
(c) 2, 2-Dimethylbutane (d) 3-Methylpentane

8. (C) Two isomeric hexyl chlorides obtained from (B) are :



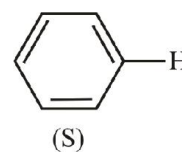
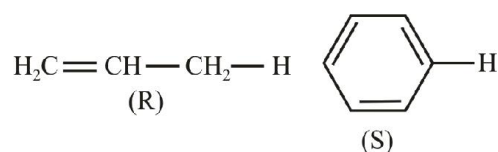
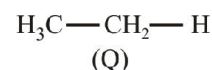
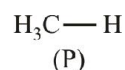
9. (C) The alkenes (C) and (D) are :



10. (S) Which of the following is incorrect regarding combustion of alkanes ?

- (a) It is always an exothermic process
(b) It proceeds through a free radical mechanism
(c) It can be controlled to an intermediate stage
(d) 2-methyl butane will liberate less heat than n-pentane.

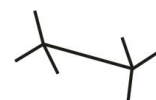
11. (S) The bond dissociation energy of the C—H bond for the compound



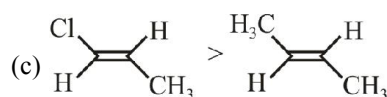
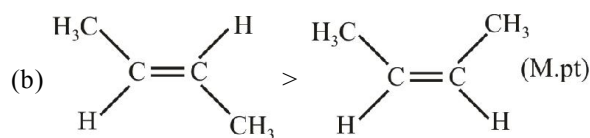
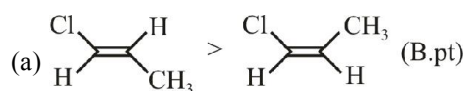
decreases in the order :

- (a) $P > Q > R > S$ (b) $S > R > Q > P$
(c) $S > P > Q > R$ (d) $Q > P > S > R$

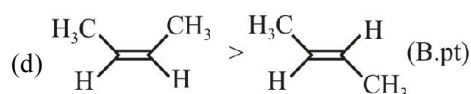
12. (I) How many mono chlorinated products may be obtained when the alkane shown below is heated in the presence of Cl_2



13. (M) Choose the correct comparisons :



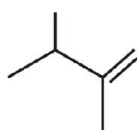
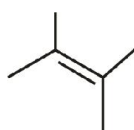
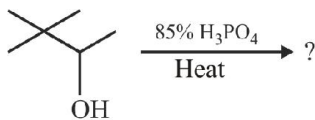
(Reactivity towards HBr)



14. (S) The boiling point of trans-dichloroethene is 48 degrees Celsius. Which number best approximates the boiling point of cis-dichloroethene ?

- (a) 36 degrees Celsius (b) 42 degrees Celsius
(c) 48 degrees Celsius (d) 54 degrees Celsius

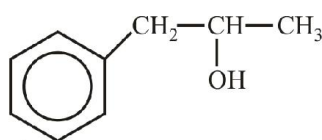
15. (S) Choose the major product of reaction



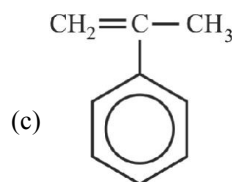
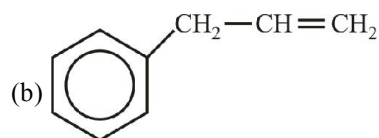
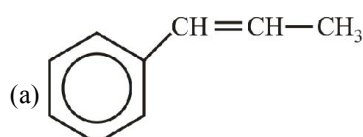
- (a) I (b) II
(c) III (d) IV

16. (S) What is the major product when 2,2-dimethylcyclohexanol is heated in acid ?

- (a) 3,3-dimethylcyclohexene
(b) 2,3-dimethylcyclohexene
(c) 2,2-dimethylcyclohexene
(d) 1,2-dimethylcyclohexene

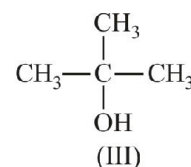


17. (S) P should be



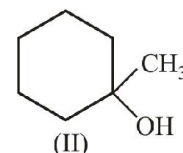
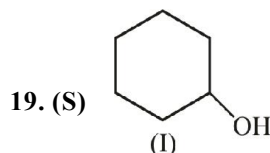
(d) none of these

18. (S) $\text{CH}_3-\text{CH}_2-\text{OH}$ (I) $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3$ (II)

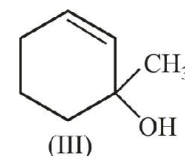


Among these alcohols, which of the following orders is correct about their ease of dehydration under acidic condition ?

- (a) I > II > III (b) III > II > I
(c) II > I > III (d) III > I > II



19. (S)

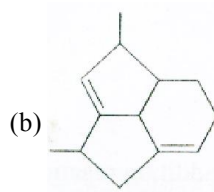
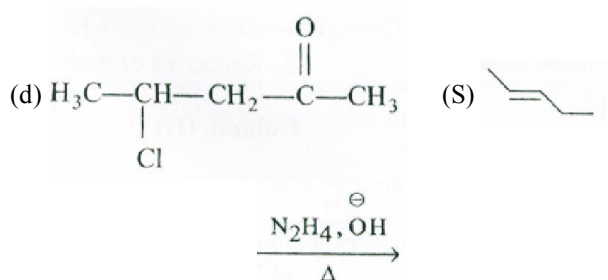
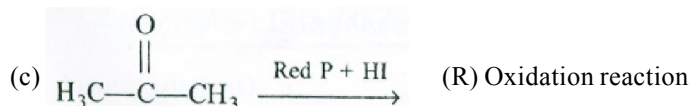
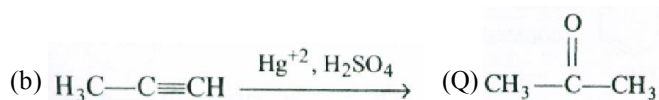
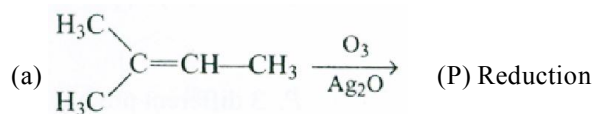
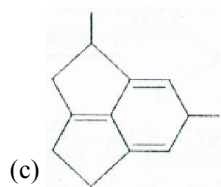
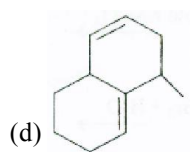


Among these alcohols, which of the following orders is correct about their ease of dehydration under acidic condition ?

- (a) I > II > III (b) III > II > I
(c) II > III > I (d) II > I > III

20. (X) Column (I)

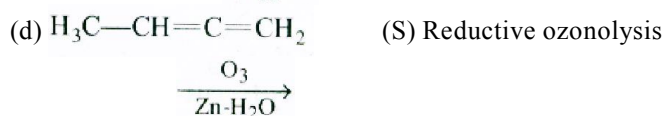
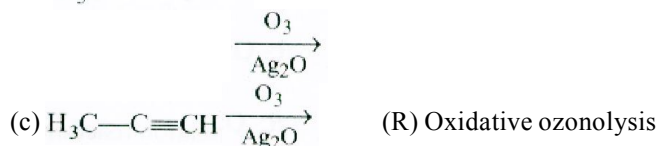
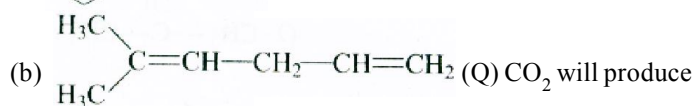
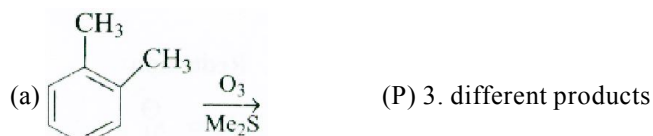
Column (II)


(Q) Decolourise $\text{Br}_2/\text{H}_2\text{O}$.

(R) Dicarboxylic acid and will be formed when reacts with $\text{O}_3/\text{H}_2\text{O}_2$.


(S) Number of allylic hydrogen is odd.

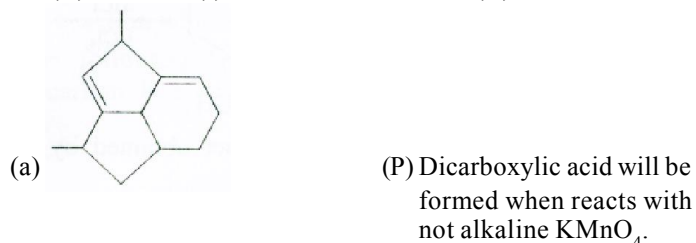
21. (X) Column (I)

Column (II)



22. (X) Column (I)

Column (II)


23. (A) **Assertion :** Alkenes usually undergo electrophilic addition reaction at carbon-carbon double bond.

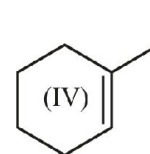
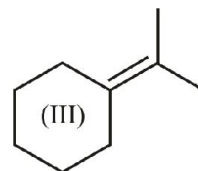
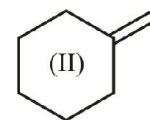
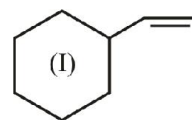
Reason : Electrophilic addition at double bond produces carbocation intermediate.

- (a) A (b) B
(c) C (d) D

24. (A) **Assertion :** A mixture of alkene and alkane can be separated by passing the mixture through a cold, concentrated sulphuric acid solution.

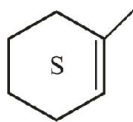
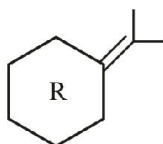
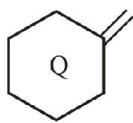
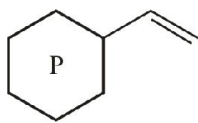
Reason : Alkene is absorbed in the solution while alkane passes unreacted.

- (a) A (b) B
(c) C (d) D

25. (S) Arrange the following alkene in increasing order of their enthalpy of hydrogenation ($-\Delta H$)


- (a) III < IV < II < I (b) III < IV < I < II
(c) I < II < III < IV (d) I < II < IV < III

26. (S) Arrange the following alkenes in increasing order of their enthalpy of hydrogenation ($-\Delta H$) :



- (a) $R < S < Q < P$ (b) $R < S < P < Q$
(c) $P < Q < R < S$ (d) $P < Q < S < R$

27. (A) **Assertion :** Hydrogenation of an alkene is always an exothermic process.

Reason : Alkane has lower potential energy than an alkene.

- (a) A (b) B
(c) C (d) D

28. (A) **Assertion :** Addition of HBr in presence of a peroxide (ROOR) to propene gives 2-bromo propane.

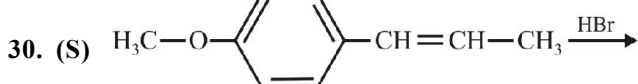
Reason : Reaction involves a bromine free radical.

- (a) A (b) B
(c) C (d) D

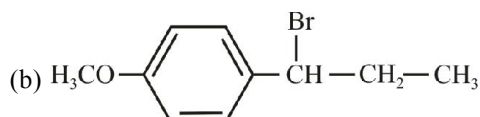
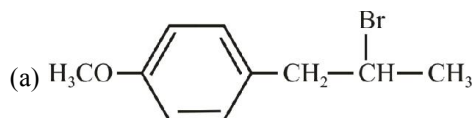
29. (S) Which of the following alkenes is most reactive towards electrophilic addition reaction ?

- (a) $H_2C = CH_2$ (b) $CH_3 - CH = CH_2$

- (c) (d) $H_2C = CH - Cl$

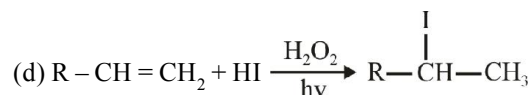
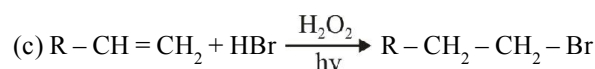
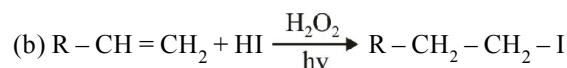
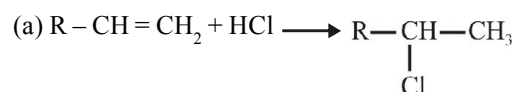


Major product :



- (c) Both (a) and (b) in same amount
(d) None of the above

31. (M) Which of the following reactions are correctly represented ?

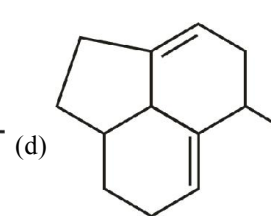
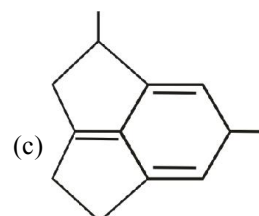
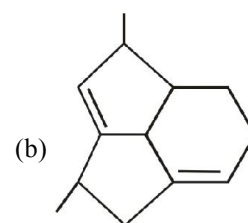
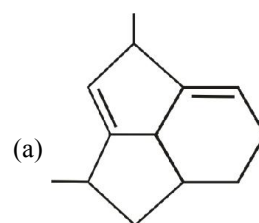
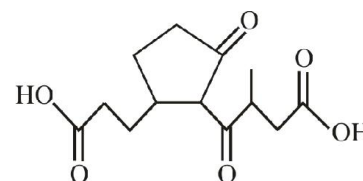


32. (A) **Assertion :** Addition of HCl to 3-methyl-1-butene gives 2-chloro-2-methyl butane as the major product.

Reason : Rearrangement of carbocation to more stable carbocation leads to formation of major product.

- (a) A (b) B
(c) C (d) D

33. (S) Which molecule will give following dicarboxylic acid upon treatment with acidic solution of $KMnO_4$?

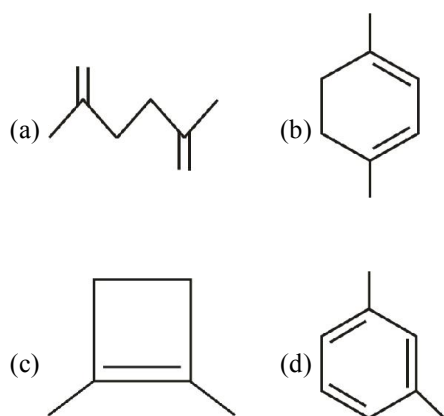
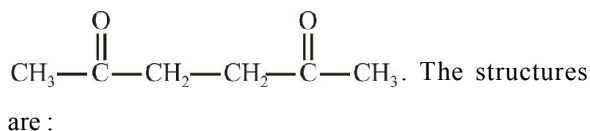


34. (A) Assertion : Treatment of one mole of 2-butene with O_3 followed by work-up with H_2O_2 gives 2 moles of acetic acid.

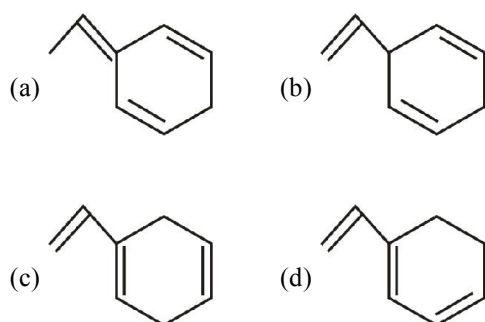
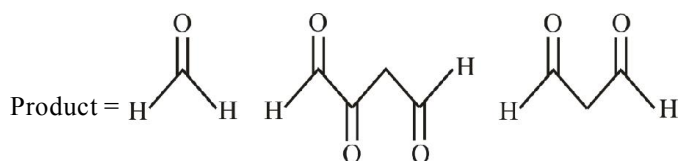
Reason : H_2O_2 bring about oxidative cleavage of intermediate ozonide.

- (a) A (b) B
(c) C (d) D

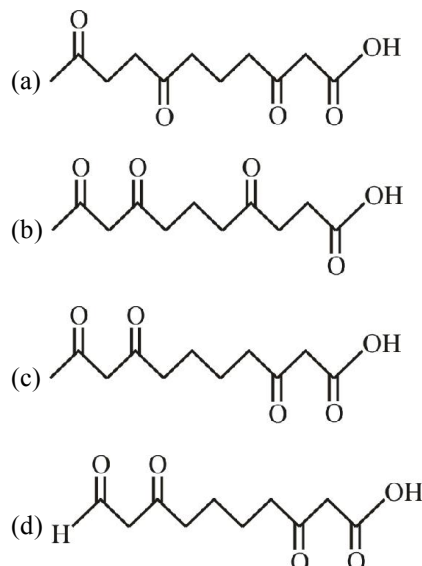
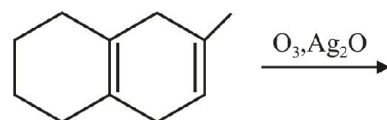
35. (M) An organic compound on reaction with O_3 followed by Zn and H_2O gives



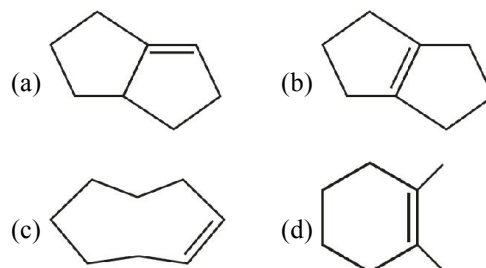
36. (S) A triene treated with ozone followed by CH_3-S-CH_3 to give following three products. What is the structure of triene ?

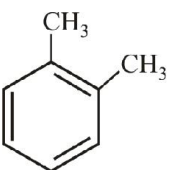


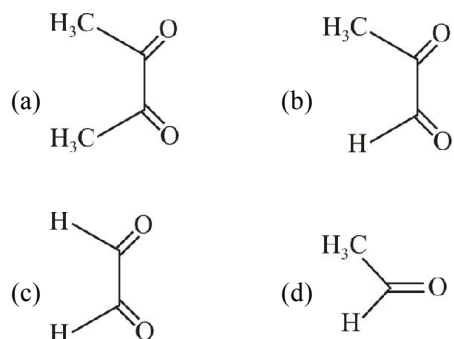
37. (S) What is the final product of the following reaction ?



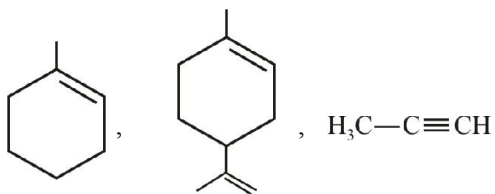
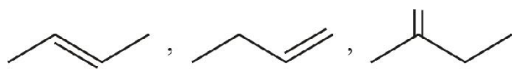
38. (S) Which of the following compounds produces-1,5-cyclooctanedione on ozonolysis ?



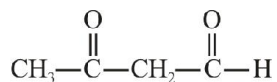
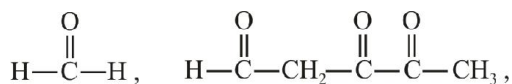
39. (M)  $\xrightarrow[Me_2S]{O_3}$ Products :



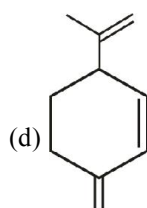
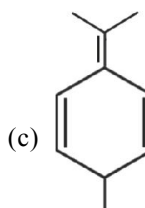
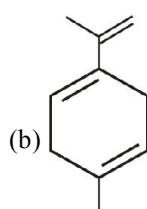
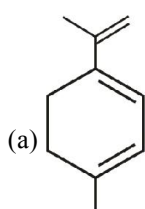
40. (I) Examine the structural formulas of following compounds and find how many compounds will produce CO_2 on oxidative ozonolysis.



41. (S) An unknown alkene (A) reacts with 3 mole of H_2 gas in presence of platinum catalyst to form 1-isopropyl-4-methyl cyclohexane. When unknown alkene (A) is ozonized and reduced, following product are obtained



The alkene (A) is



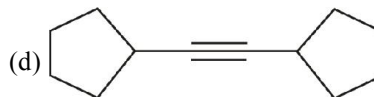
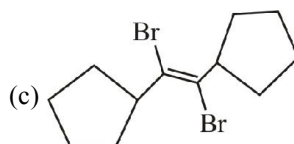
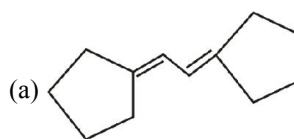
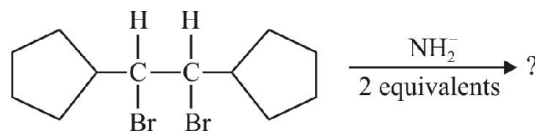
42. (M) Which of the statements are correct for alkyne with molecular formula C_6H_{10} ?

- (a) It shows 7 structural isomers.
(b) It shows 4 terminal and 3 internal alkynes.
(c) It shows 3 terminal and 4 internal alkynes.
(d) Only one isomer is chiral.

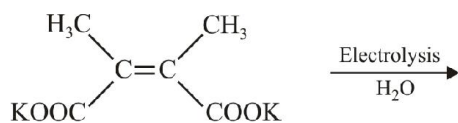
43. (S) What is the relationship between the two terminal $\text{H}_2\text{C} =$ groups in a cumulene containing three consecutive double bonds ($\text{H}_2\text{C}=\text{C}=\text{C}=\text{CH}_2$) ?

- (a) They are in the perpendicular plane
(b) They are in the same plane
(c) They are in two planes at 120°
(d) They are in two planes at 109°

44. (S) What is the major product in the reaction ?



46. (S) What will be formed in the following Kolbe's electrolysis ?



- (a) 2-butene (b) 1-butyne
(c) 2-butyne (d) Both b and c
47. (M) Which of the following will produce a visible change with 1-butyne and can be used to differentiate it from 2-butyne ?

- (a) $\text{Ag}(\text{NH}_3)_2^+$ (b) $\text{Cu}(\text{NH}_3)_2^+$
(c) KMnO_4/H^+ (d) Br_2

48. (A) **Assertion :** 1-butyne forms white precipitate with ammoniacal silver nitrate solution while 2-butyne does not.

Reason : 1-butyne has acidic hydrogen.

- (a) A (b) B
(c) C (d) D

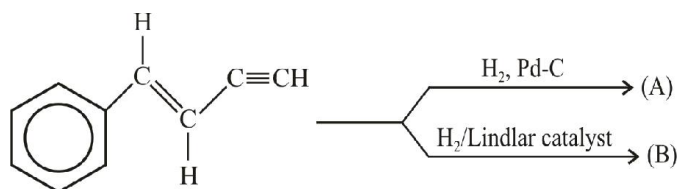
49. (M) Which of the following reactions are not feasible ?

- (a) $\text{HC}\equiv\text{CH} + \text{KOH} \longrightarrow$
(b) $\text{HC}\equiv\text{CH} + \text{NaNH}_2 \longrightarrow$
(c) $\text{HC}\equiv\text{CH} + \text{NaOH} \longrightarrow$
(d) $\text{HC}\equiv\text{C}^-\text{K}^+ + (\text{CH}_3)_3\text{C}-\text{Br} \longrightarrow$

50. (M) Which of the following will react with 1-butyne ?

- (a) $\text{AgNO}_3 + \text{NH}_4\text{OH}$ (b) $\text{Cu}_2\text{Cl}_2 + \text{NH}_4\text{OH}$
(c) Na (d) $\text{KMnO}_4/\text{OH}^-$

51. (M) Predict the products of following reactions :



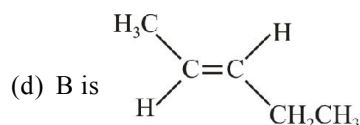
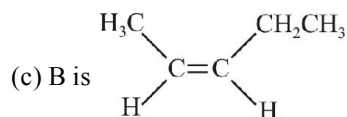
- (a) A is $\text{Ph}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$
(b) B is $\text{Ph}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$

(c) A is $\text{Ph}-\text{CH}=\text{CH}-\text{CH}=\text{CH}_2$

(d) B is $\text{Ph}-\text{CH}_2-\text{CH}_2-\text{CH}=\text{CH}_2$

52. (M) $\text{H}_3\text{C}-\text{C}\equiv\text{CH} \xrightarrow[2. \text{CH}_3\text{CH}_2\text{Br}]{1. \text{NaNH}_2} \text{A} \xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2} \text{B} :$

- (a) A is $\text{H}_3\text{C}-\text{CH}_2-\text{C}\equiv\text{CH}$
(b) A is $\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_2\text{CH}_3$



53. (A) **Assertion :** 2-butyne on controlled hydrogenation with Pd/CaCO_3 in presence of PbO gives cis-2-butene.

Reason : Hydrogenation occur at the surface of metal containing adsorbed hydrogen.

- (a) A (b) B
(c) C (d) D

54. (A) **Assertion :** Propyne is more reactive than propene towards electrophilic addition of HBr .

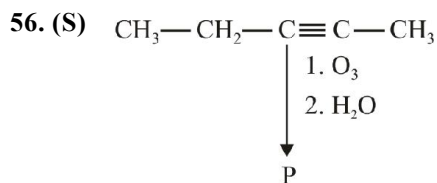
Reason : Propyne has greater electron density available for electrophile due to two pi-bonds, therefore more electrophilic.

- (a) A (b) B
(c) C (d) D

55. (A) **Assertion :** Ozonolysis of 3-hexyne followed by hydrolysis gives propanoic acid.

Reason : The byproduct H_2O_2 oxidizes ozonide to acid.

- (a) A (b) B
(c) C (d) D



P should be

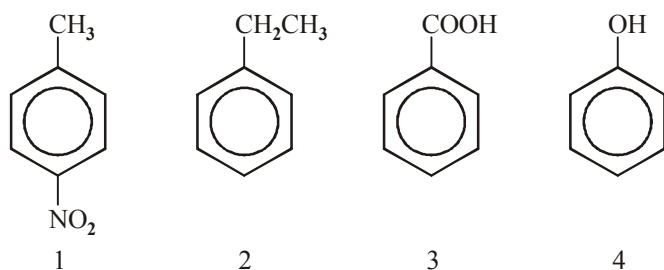
- (a) $\text{CH}_3-\text{CH}_2-\text{CHO} + \text{CH}_3\text{CHO}$
(b) $\text{CH}_3-\text{CH}_2-\text{COOH} + \text{CH}_3\text{COOH}$

- (c) $\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{CH}_3$
(d) none of these

57. (S) Which of the following cannot be used to prepare cumene from benzene ?

- (a) $\text{CH}_3-\text{CH}=\text{CH}_2$, HF
(b) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl}$, AlCl_3
(c) $\text{CH}_2=\text{CH}_2$, AlCl_3
(d) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$, BF_3

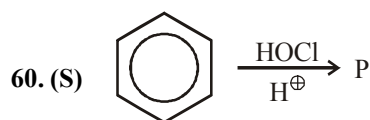
58. (S) Which will undergo Friedal-Craft alkylation reaction ?



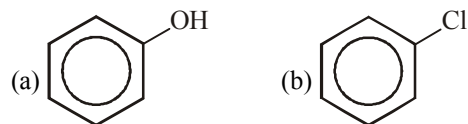
- (a) 1 and 3 (b) 2 and 4
(c) 1 and 2 (d) 1, 2 and 4

59. (S) Nitration of benzene is possible by which of the following reagents ?

- (a) Conc. HNO_3 /Conc. H_2SO_4
(b) $\text{R}-\text{NO}_2/\text{H}^+$
(c) both (a) and (b)
(d) none of these

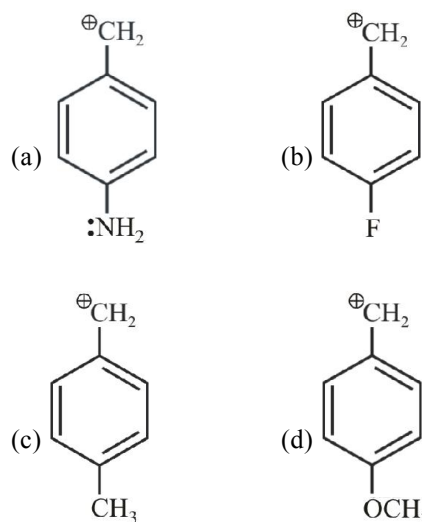


Based on the above reaction, P should be

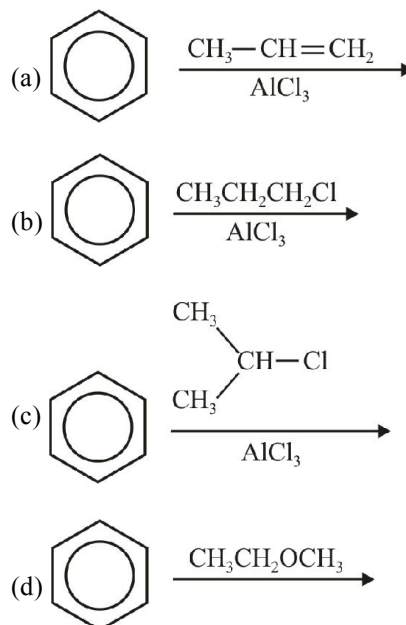


(c) mixture of (a) and (b) (d) none of these

61. (M) Which of the following cations are more stable than benzyl cation ?

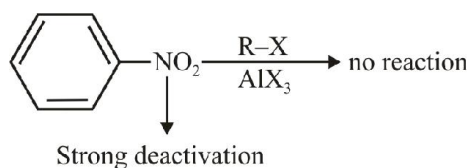


62. (M) Isopropyl benzene can be obtained by :

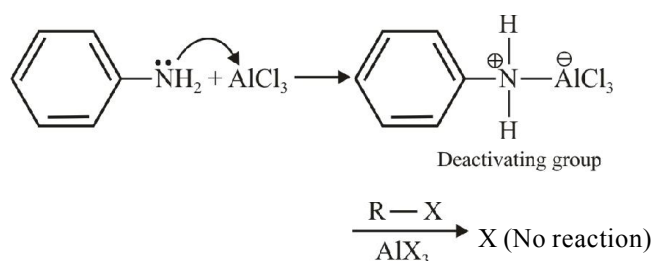


Passage

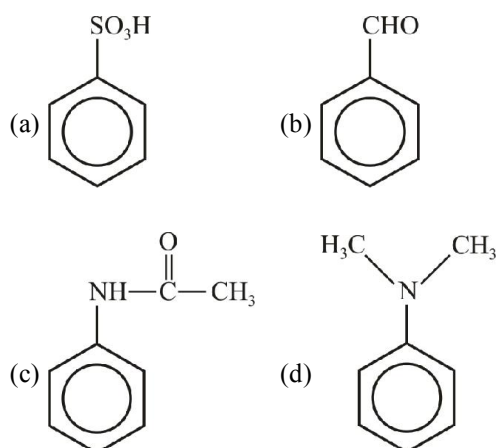
A benzene ring deactivated by strong and moderate electron withdrawing group that is, any meta directing group, is not electron rich enough to undergo Friedel-Crafts reactions.



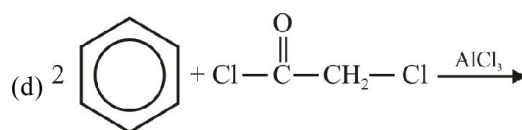
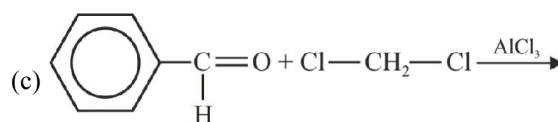
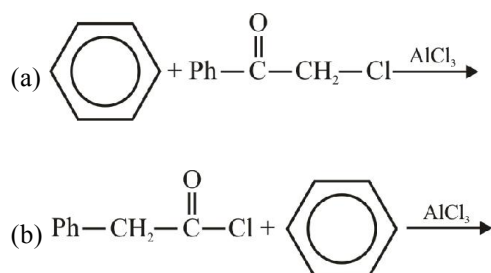
Friedel-Crafts reaction also do not occur with NH_2 group as it react with AlCl_3 and produce deactivating group.



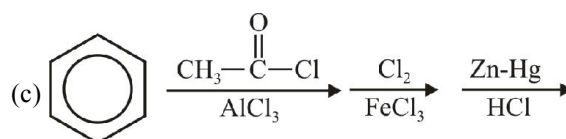
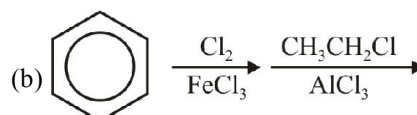
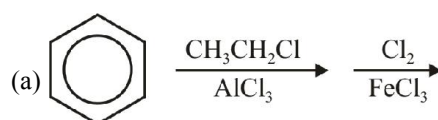
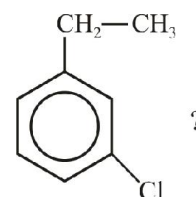
62. (C) Which of the following compounds undergo Friedel-Crafts alkylation reaction ?



63. (C) Which of the following cannot be starting material for this compound $\text{Ph}-\text{C}(=\text{O})-\text{CH}_2-\text{Ph}$?

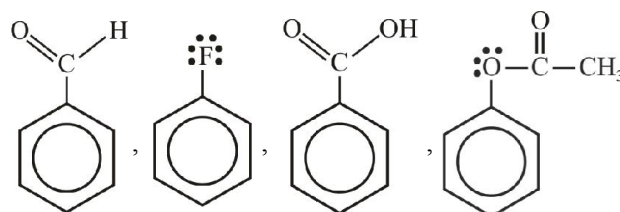
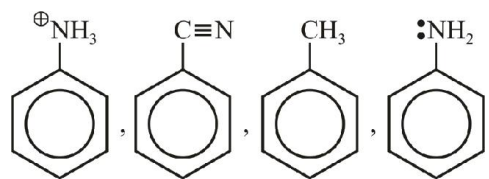


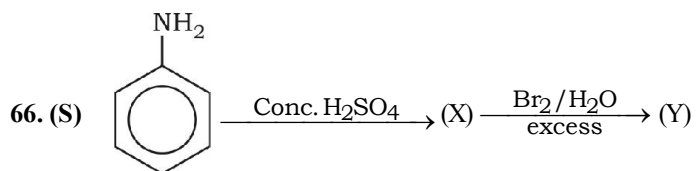
64. (C) Which of the following sequence of reaction is correct for the synthesis of product



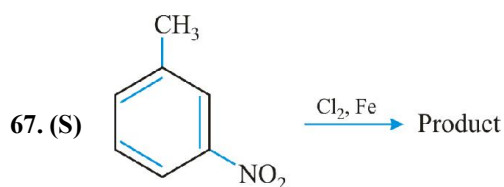
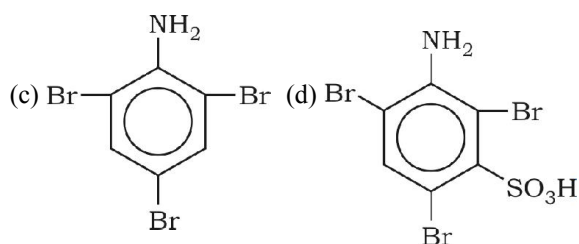
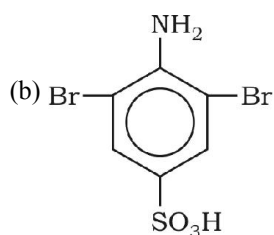
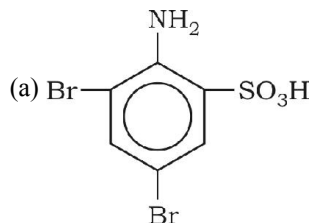
(d) All are incorrect

65. (I) Examine the structural formulas shown below and find out how many compounds can not, give Friedel Craft reaction.

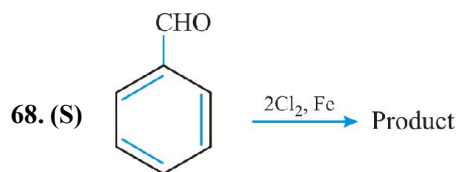
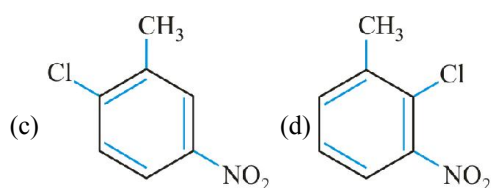
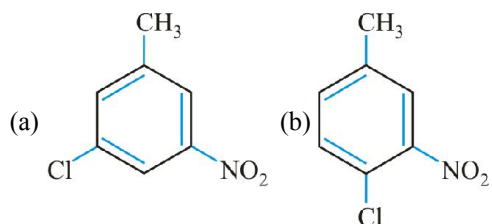




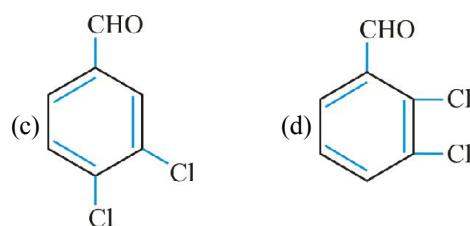
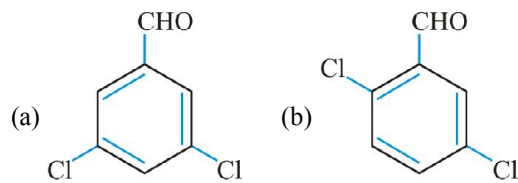
: Product (Y) of this reaction is



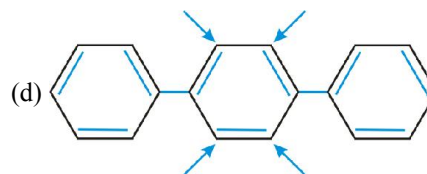
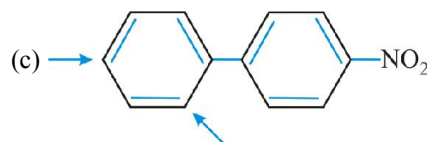
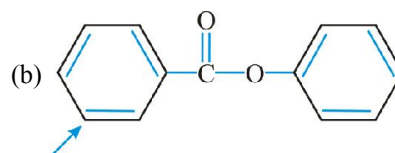
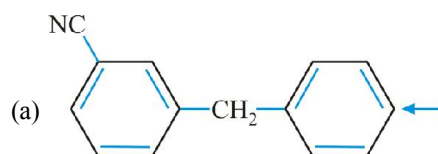
The predominant product is

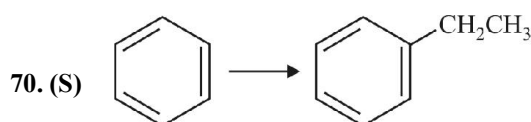


The final predominant product is

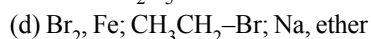
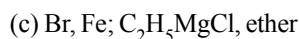
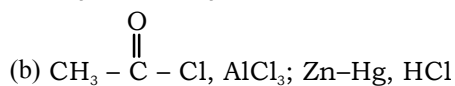


69. (S) In which of the following the incorrect place for electrophilic substitution has been indicated ?

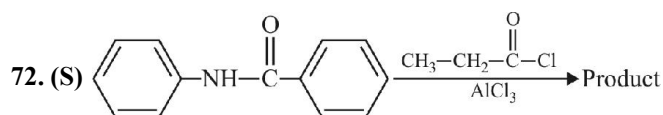
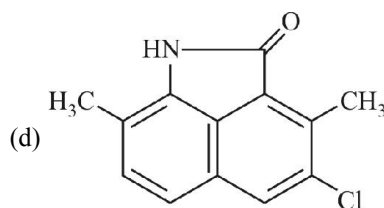
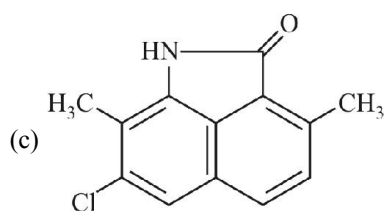
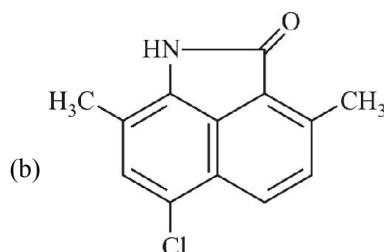
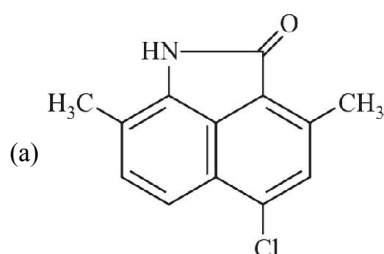
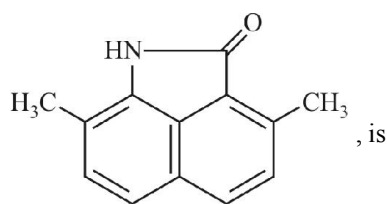




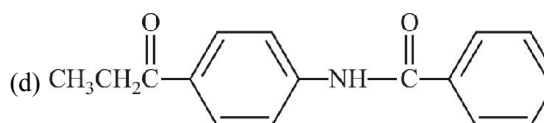
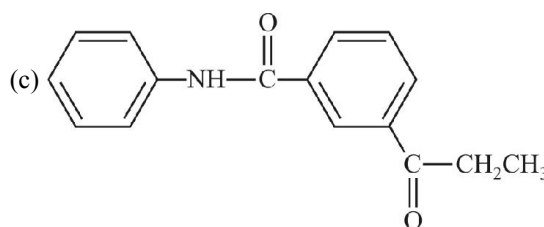
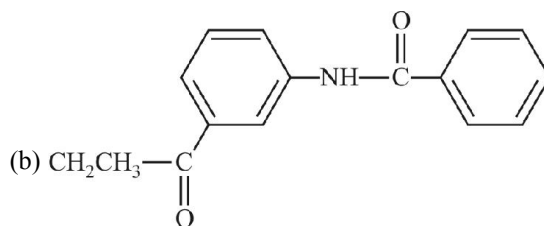
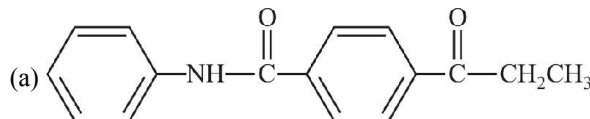
Which of the following is not the appropriate way of conversion ?



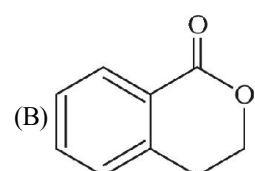
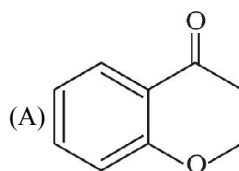
71. (S) Product of monochlorination of the compound



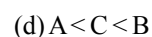
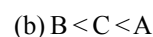
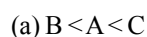
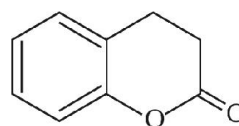
Which of the following is the correct structure of the organic product of the above reaction ?



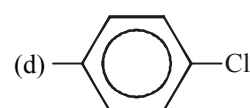
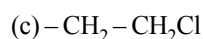
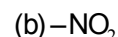
73. (S) Rank in order of increasing rate of reaction towards EAS with bromine in the presence of FeBr_3 .



(C)



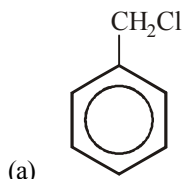
74. (M) Which of the following is/are deactivating groups towards electrophilic aromatic substitution ?



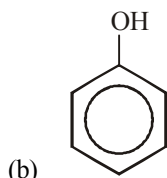
75. (X) Match the following :

Column-I

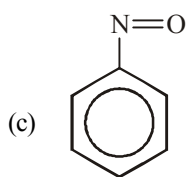
Column-II



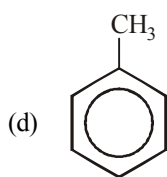
(p) Activated towards electrophilic aromatic substitution



(q) Deactivated towards electrophilic aromatic substitution



(r) Ortho/Para director



(s) Meta-director

76. (A) **Assertion :** Toluene undergoes nitration much more readily than benzene.

Reason : It is due to electron releasing nature of $-\text{CH}_3$ group which increases electron density on benzene and electrophilic substitution reaction like nitration becomes faster.

(a) A

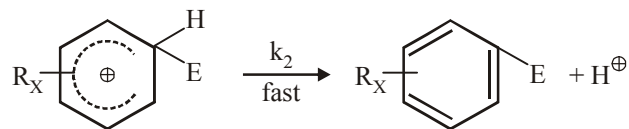
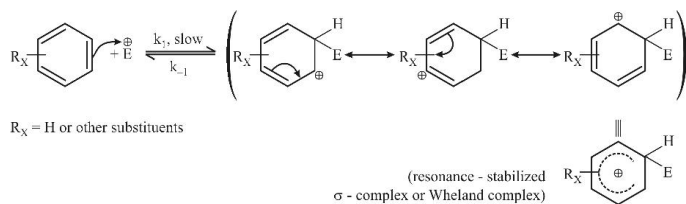
(b) B

(c) C

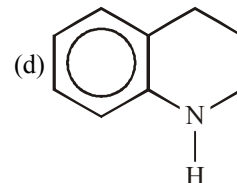
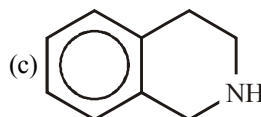
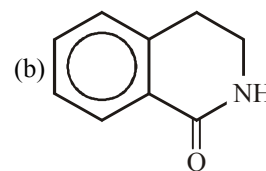
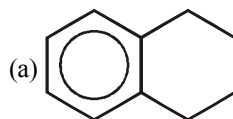
(d) D

Passage

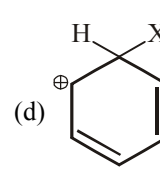
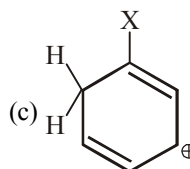
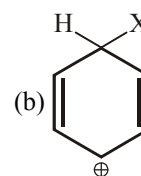
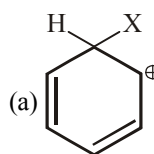
Aromatic compounds can easily show electrophilic substitution reactions due to the presence of π electron clouds above and below the plane of aromatic ring. This ring can be easily attacked by electrophiles due to high electro density. The mechanism of these reaction is common in general which can be given as follows :



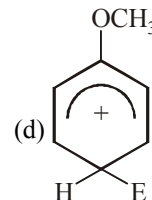
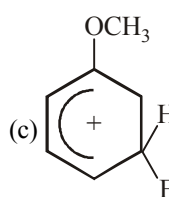
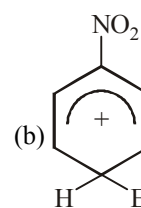
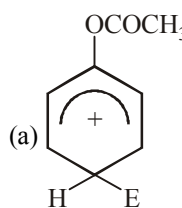
77. (C) In which of the following compounds the rate of electrophilic substitution (Halogenation) is fastest ?



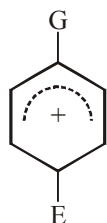
78. (C) Which of the following resonance structures is not a contributor to the cyclo hexa dienyl cation intermediate in the halogenation of benzene ?



79. (C) Which of the following σ -complexes generated by the attack of an electrophile of benzene ring has lowest energy ?



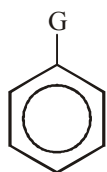
80. (I) Mark the following in increasing order of stability for different Gs for the following Wheland complex



- (1) CH_3 (2) NO_2
 (3) OCH_3 (4) —NH—C(=O)—CH_3

81. (I) For the following reactions of substituted benzene, mark the answer using the following guide line

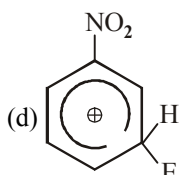
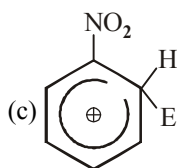
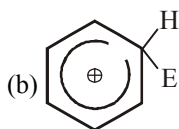
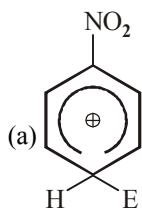
- (i) If substitution takes place at ortho/para position mark 1.
 (ii) If substitution takes place at meta, mark 2.
 (iii) If reaction takes place at position other than these, mark 3.
 (iv) In case of no reaction, mark 4.



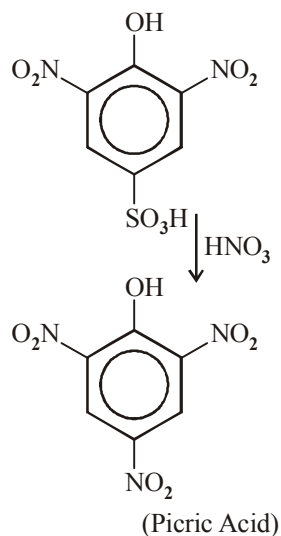
G = substituent

- (A) $\text{G}=\text{CH}_3$. Reaction with CH_3Cl , AlCl_3 .
 (B) $\text{G}=\text{CH}_3$. Reaction with Cl_2 , $h\nu$.
 (C) $\text{G}=\text{NO}$. Reaction with Cl_2 , Fe .
 (D) $\text{G}=\text{NO}_2$. Reaction with CH_3Cl , AlCl_3 .

82. (S) An electrophile, E^+ attacks the benzene ring to generate the intermediate σ -complex. Of the following, which σ -complex is of lowest energy?



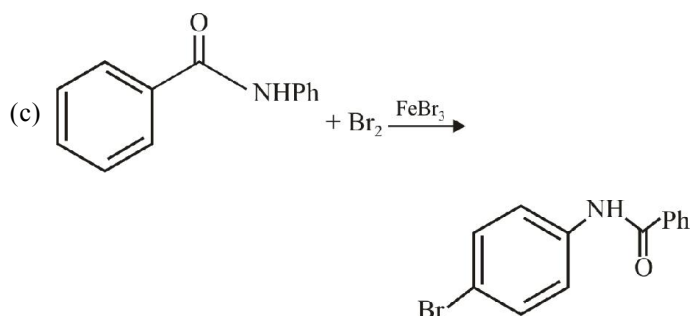
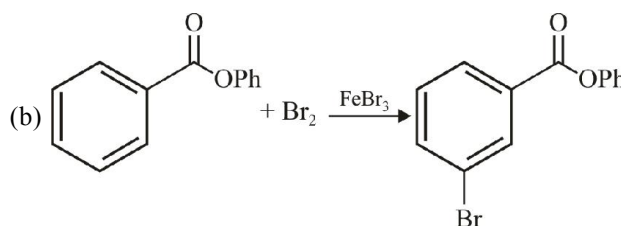
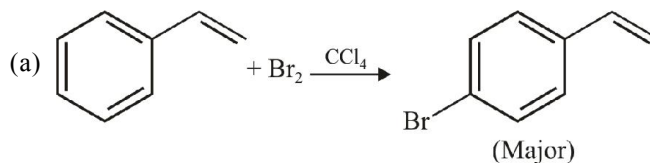
83. (S) The sulfonic acid group is often readily replaced by a nitro group e.g.,

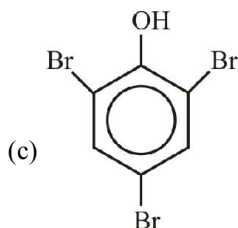
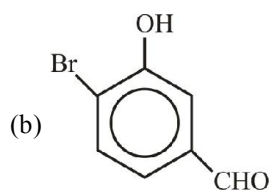
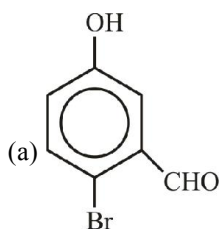
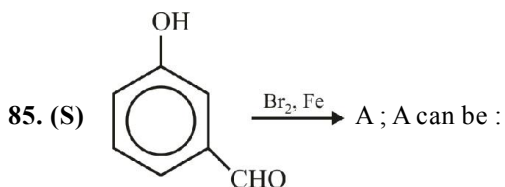
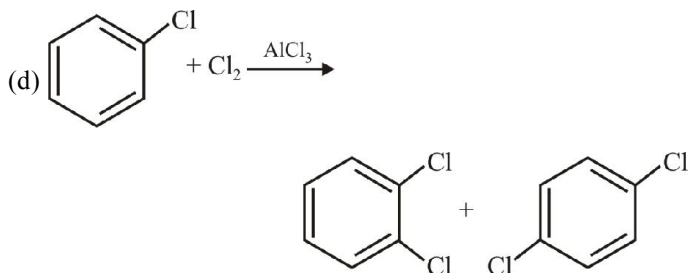


This is because

- (a) $\text{—SO}_3\text{H}$ group is a better leaving group than —NO_2 group in electrophilic substitution
 (b) —NO_2 group is a better leaving group than $\text{—SO}_3\text{H}$ group in electrophilic substitutions
 (c) This is a nucleophilic substitution.
 (d) both (b) and (c)

84. (M) In which of the followings, reactant and product are correctly matched?



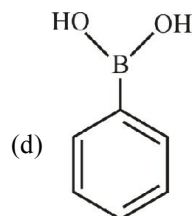
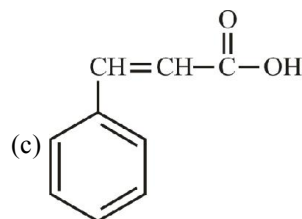
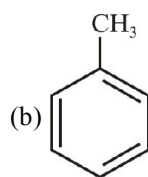
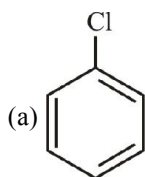


(d) no reaction

86. (M) Halogens are deactivating yet ortho, para directing in electrophilic aromatic substitution. Which statements do not explain this ?

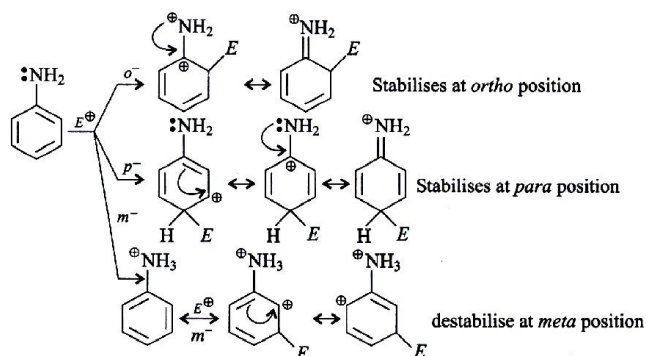
- A combination of inductive electron withdrawal and resonance electron release.
- Inductive electron withdrawal with no resonance effect.
- A combination of inductive electron release and resonance electron withdrawal.
- A combination of inductive electron release and resonance electron release

87. (M) Which of the following are deactivating but ortho, para directing during electrophilic aromatic substitution reaction ?



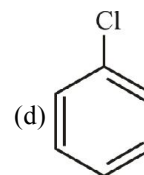
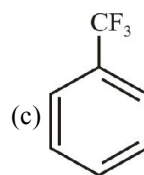
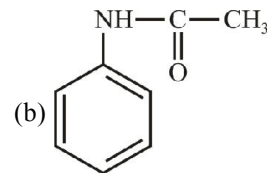
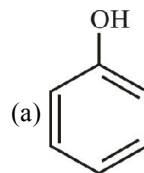
Passage

Directing nature of substituted aromatic compound is decided by stability of σ -complex or arenium ion. If σ -complex is stabilise at o-and p-position by attack of electrophile then the group is o-and p-directing but if σ -complex is stabilise at m-position then group will be meta directing

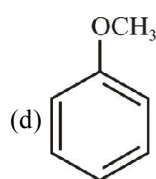
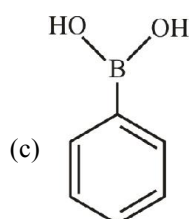
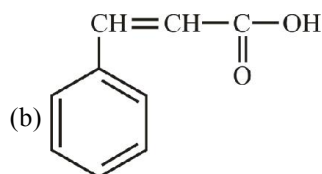
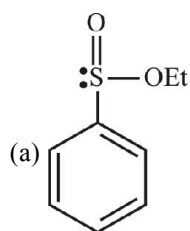


On the basis of above explanation find out correct answers of following questions.

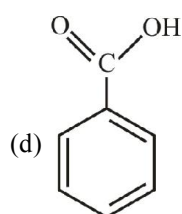
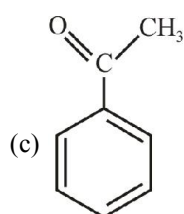
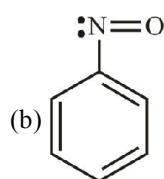
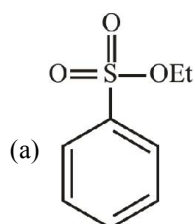
88. (C) Which of the following is m-directing ?



89. (C) Which of the following is not o- and p-directing ?

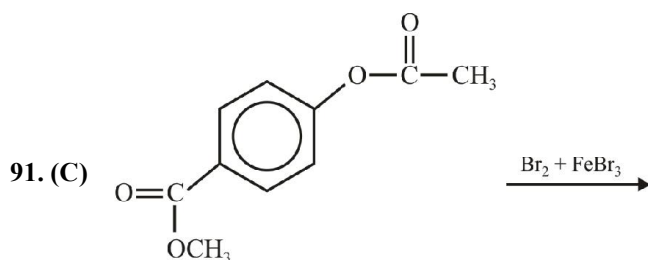


90. (C) Which of the following is o- and p-directing ?

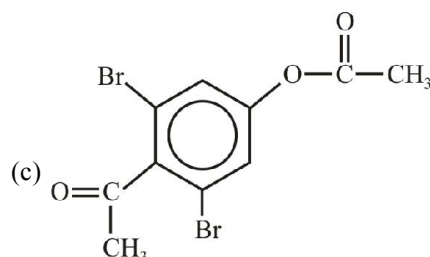
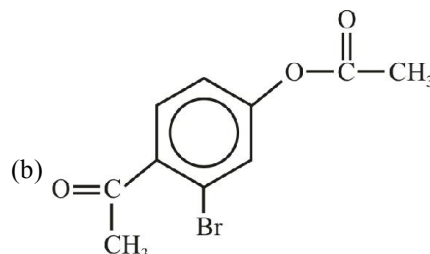
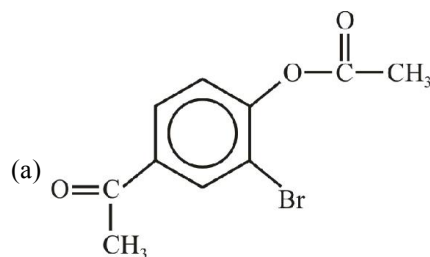


Passage

If aromatic ring is substituted by more than groups then electrophilic aromatic substitution reaction take place according to more activating group. Types of group which donate electron in aromatic ring known as activating groups.

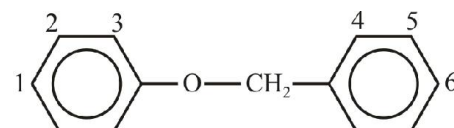


Find out correct product of reaction.



(d) No reaction

92. (C) Major product formation take place at which position in this reaction :



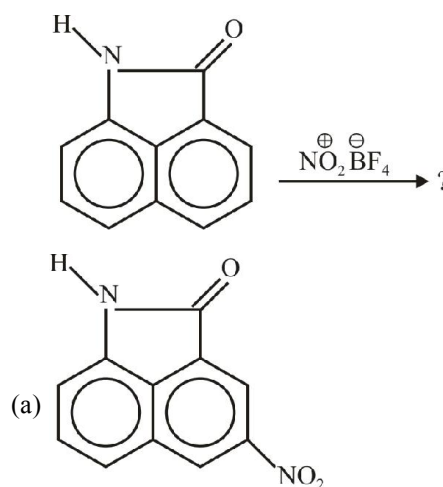
(a) 3

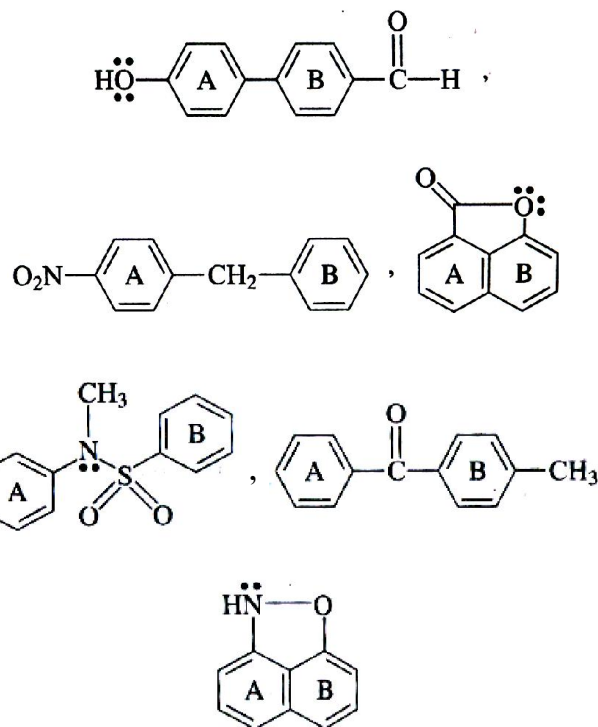
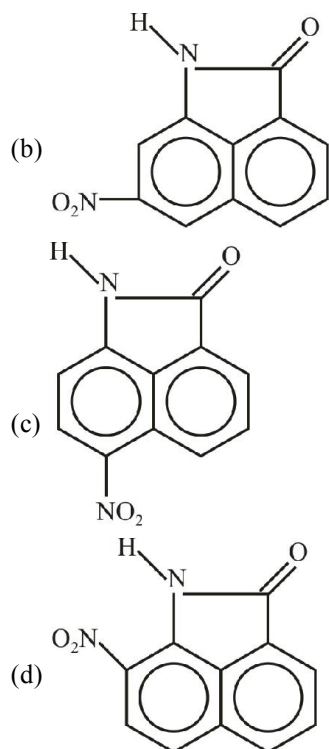
(b) 1

(c) 6

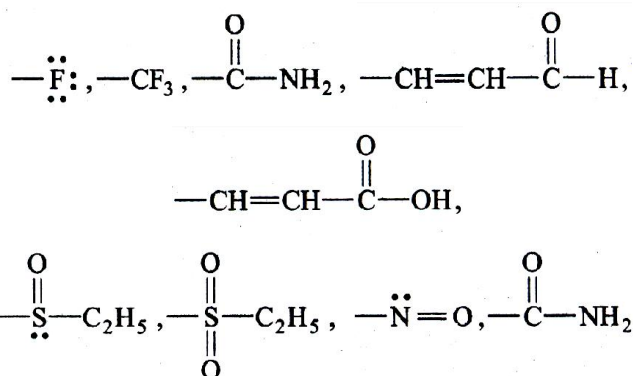
(d) 4

93. (C) Find out major product of following reaction :

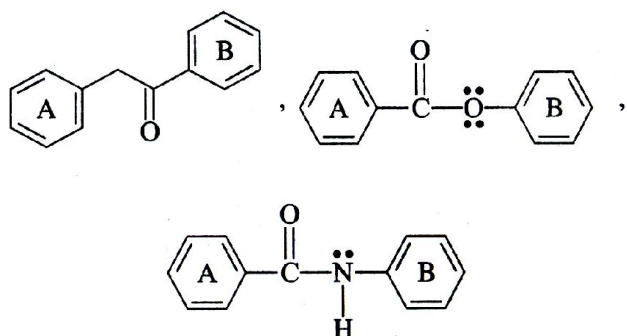




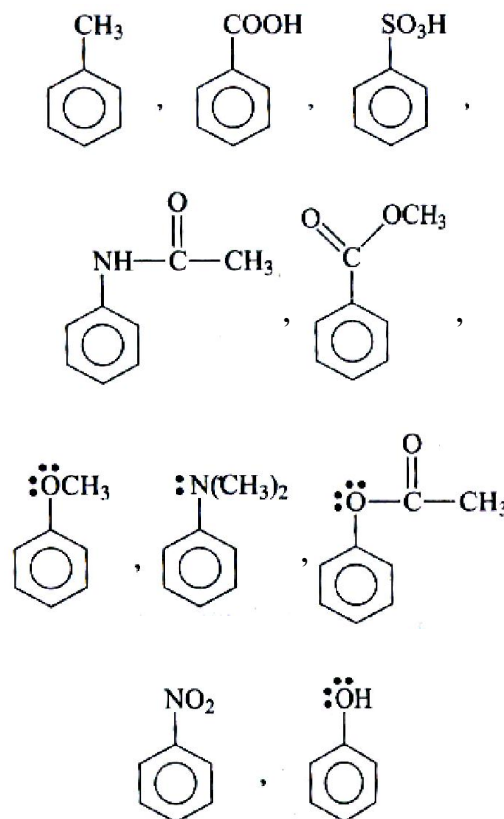
94. (I) Identify number of substituents those are deactivating but ortho and para directing.



95. (I) Each of the compounds shown below has two aromatic ring, labeled as A and B. Identify number of compounds in which ring B is more active than ring A for electrophilic aromatic substitution reaction.



96. (I) Examine the structural formulas shown below and find out how many compounds undergo electrophilic nitration more rapidly than fluoro benzene.

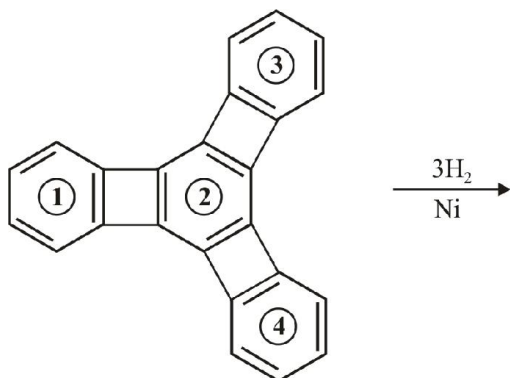


97. (A) **Assertion** : Benzene does not decolourise alkaline KMnO_4 .

Reason : Benzene is stabilized by resonance and π -electrons are delocalized.

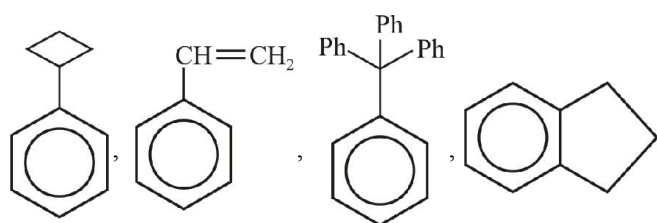
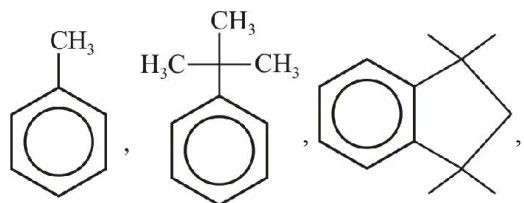
- (a) A (b) B
(c) C (d) D

98. (I)

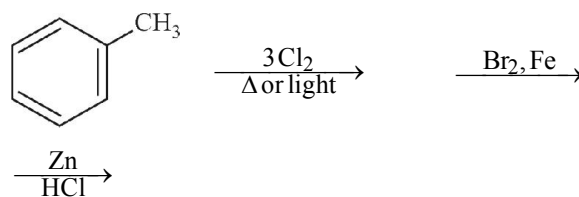


In this reaction which ring will be reduced, by hydrogenation.

99. (I) Examine the structural formulas shown below and find out how many compounds will show oxidation reaction with acidic KMnO_4 .



100. (S)



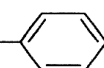
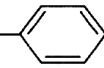
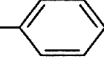
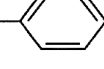
Product

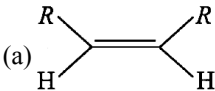
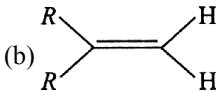
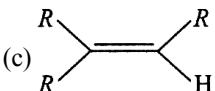
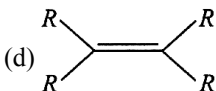
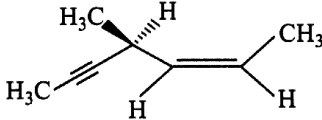
The main product is

- (a)
- (b)
- (c)
- (d)

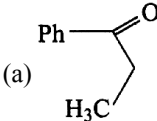
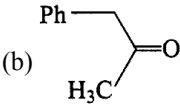
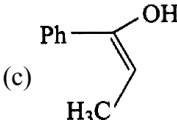
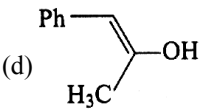
EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTIONS

Objective Questions

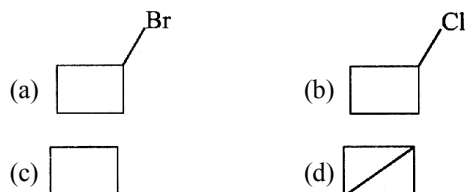
1. Marsh gas mainly contains (1980)
(a) C_2H_2 (b) CH_4
(c) H_2S (d) CO
2. Which of the following will decolourise alkaline $KMnO_4$ solution? (1980)
(a) C_3H_8 (b) CH_4
(c) CCl_4 (d) C_2H_4
3. The compound with highest boiling point is (1982)
(a) 2-methyl butane (b) n-pentane
(c) 2,2-dimethyl propane (d) n-hexane
4. The compound 1, 2-butadiene has (1983)
(a) only sp -hybridised carbon atoms
(b) only sp^2 -hybridised carbon atoms
(c) both sp and sp^2 - hybridised carbon atoms
(d) sp , sp^2 and sp^3 -hybridised carbon atoms
5. When propyne is treated with aqueous H_2SO_4 in presence of $HgSO_4$, the major product is (1983)
(a) propanal
(b) propyl hydrogen sulphate
(c) acetone
(d) propanol
6. Which of the following compounds does not dissolve in conc. H_2SO_4 , even on warming? (1983)
(a) Ethylene (b) Benzene
(c) Hexane (d) Aniline
7. Bayer's reagent is (1984)
(a) alkaline permanganate solution
(b) acidified permanganate solution
(c) neutral permanganate solution
(d) aqueous bromine solution
8. Acidic hydrogen is present in (1985)
(a) ethyne (b) ethene
(c) benzene (d) ethane
9. The highest boiling point is expected for (1986)
(a) iso-butane (b) n-octane
(c) 2, 2, 3, 3-tetramethyl butane
(d) 2-methyl pentane
10. The number of structural and configurational isomers of a bromo compound, C_5H_9Br , formed by the addition of HBr to 2-pentyne respectively, are (1988)
(a) 1 and 2 (b) 2 and 4
(c) 4 and 2 (d) 2 and 1
11. The compound which has one isopropyl group, is (1989)
(a) 2, 2, 3, 3-tetramethyl pentane
(b) 2, 2-dimethyl pentane
(c) 2, 2, 3- trimethyl pentane (d) 2-methyl pentane
12. The C—H bond distance is the longest in (1989)
(a) C_2H_2 (b) C_2H_4
(c) C_2H_6 (d) $C_2H_2Br_2$
13. Which one of the following has the smallest heat of hydrogenation per mole? (1993)
(a) 1-butene (b) trans-2-butene
(c) cis-2-butene (d) 1, 3-butadiene
14. When cyclohexane is poured on water, it floats, because (1997)
(a) cyclohexane is in 'boat' form
(b) cyclohexane is in 'chair' form
(c) cyclohexane is in 'crown' form
(d) cyclohexane is less dense than water
15. $(CH_3)_3CMgCl$ on reaction with D_2O produces (1997)
(a) $(CH_3)_3CD$ (b) $(CH_3)_3OD$
(c) $(CD_3)_3CD$ (d) $(CD_3)_3OD$
16. The reaction of $CH_3CH=CH-$  $-OH$ with HBr gives (1998)
(a) $CH_3CHBrCH_2-$  $-OH$
(b) $CH_3CHBrCH_2-$  $-Br$
(c) CH_3CH_2CHBr-  $-OH$
(d) CH_3CH_2CHBr-  $-Br$

17. Benzyl chloride ($C_6H_5CH_2Cl$) can be prepared from toluene by chlorination with (1998)
 (a) SO_2Cl_2 (b) $SOCl_2$
 (c) Cl_2 (d) $NaOCl$
18. In the compound, $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_2\text{CH}_3$, the C2-C3 bond is of the type (1999)
 (a) $sp - sp^2$ (b) $sp^3 - sp^3$
 (c) $sp - sp^3$ (d) $sp^2 - sp^3$
19. The product(s) obtained via oxymercuration ($HgSO_4 + H_2SO_4$) of 1-butyne would be (1999)
 (a) $CH_3-CH_2-C(=O)-CH_3$
 (b) $CH_3-CH_2-CH_2-CHO$
 (c) $CH_3-CH_2-CHO + HCHO$
 (d) $CH_3-CH_2-COOH + HCOOH$
20. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition? (2000)
 (a)  (b) 
 (c)  (d) 
21. Propyne and propene can be distinguished by (2000)
 (a) conc. H_2SO_4 (b) Br_2 in CCl_4
 (c) dil. $KMnO_4$ (d) $AgNO_3$ in ammonia
22. Hydrogenation of the adjoining compound in the presence of poisoned palladium catalyst gives (2001)

 (a) an optically active compound
 (b) an optically inactive compound
 (c) a racemic mixture
 (d) a diastereomeric mixture
23. The reaction of propene with $HOCl$ proceeds via the addition of (2001)
 (a) H^+ in the first step (b) Cl^+ in the first step
 (c) OH^- in the first step (d) Cl^+ and OH^- single step
24. In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti-Markownikoff's addition to alkenes because (2001)
 (a) both are highly ionic
 (b) one is oxidising and the other is reducing
 (c) one of the steps is endothermic in both the cases
 (d) all the steps are exothermic in both the cases
25. Identify a reagent from the following list which can easily distinguish between 1-butyne and 2-butyne. (2002)
 (a) bromine, CCl_4 (b) H_2 , Lindlar catalyst
 (c) dilute H_2SO_4 , $HgSO_4$ (d) ammoniacal $CuCl_2$ solution
26. Consider the following reaction

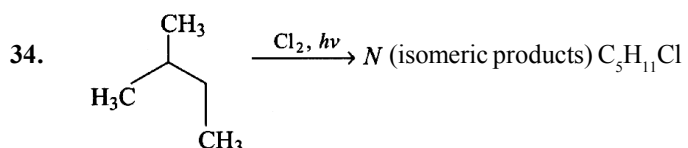
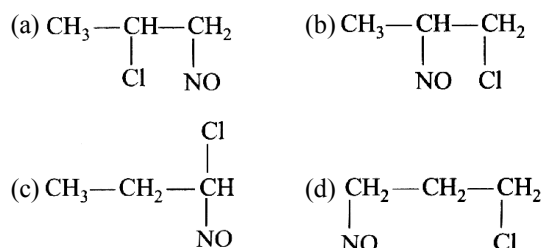
$$H_3C-\underset{\substack{| \\ D}}{CH}-\underset{\substack{| \\ CH_3}}{CH}-CH_3 + \dot{Br} \longrightarrow 'X' + HBr$$

 Identify the structure of the major product X (2002)
 (a) $H_3C-\underset{\substack{| \\ D}}{CH}-\underset{\substack{| \\ CH_3}}{CH}-\dot{C}H_2$
 (b) $H_3C-\underset{\substack{| \\ D}}{CH}-\dot{C}-CH_3$
 (c) $H_3C-\dot{C}-\underset{\substack{| \\ CH_3}}{CH}-CH_3$
 (d) $H_3C-\dot{C}H-\underset{\substack{| \\ CH_3}}{CH}-CH_3$
27. $Ph-C\equiv C-CH_3 \xrightarrow[H^+]{Hg^{2+}} A$; A is (2003)
 (a)  (b) 
 (c)  (d) 
28. 2-phenyl propene on acidic hydration, gives (2004)
 (a) 2-phenyl-2-propanol (b) 2-phenyl-1-propanol
 (c) 3-phenyl-1-propanol (d) 1-phenyl-2-propanol

29. 2-hexyne gives trans-2 hexene on treatment with (2004)
 (a) Li/NH_3 (b) Pd/BaSO_4
 (c) LiAlH_4 (d) Pt/H_2
30. How many chiral compounds are possible on mono chlorination of 2 -methyl butane? (2004)
 (a) 2 (b) 4
 (c) 6 (d) 8
31. 1-bromo-3-chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed? (2005)

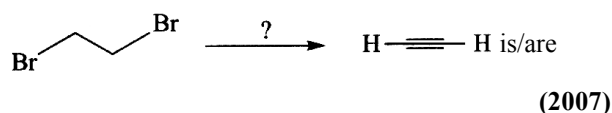


32. Cyclohexene is best prepared from cyclohexanol by which of the following? (2005)
 (a) conc. H_3PO_4 (b) conc. HCl/ZnCl_2
 (c) conc. HCl (d) conc. HBr
33. $\text{CH}_3-\text{CH}=\text{CH}_2 + \text{NOCl} \longrightarrow \text{P}$; Identify the adduct. (2006)



Fractional distillation $\longrightarrow$ M (isomeric products) What are N and M? (2006)

- (a) 6, 6 (b) 6, 4
 (c) 4, 4 (d) 3, 3
35. The reagent(s) for the following conversion,



- (a) alcoholic KOH
 (b) alcoholic KOH followed by NaNH_2
 (c) aqueous KOH followed by NaNH_2
 (d) $\text{Zn/CH}_3\text{OH}$

36. The number of stereoisomers obtained by bromination of trans-2-butene is (2007)
 (a) 1 (b) 2
 (c) 3 (d) 4
37. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are (2010)
 (a) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$
 (b) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$
 (c) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}\equiv\text{CH}$
 (d) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

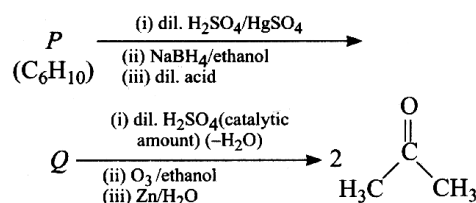
Assertion and Reason

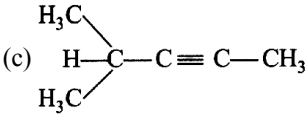
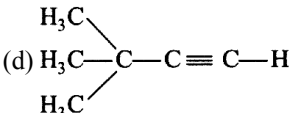
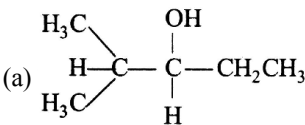
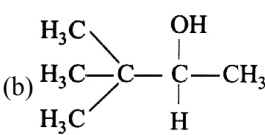
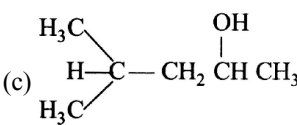
Read the following questions and answer as per the direction given below:

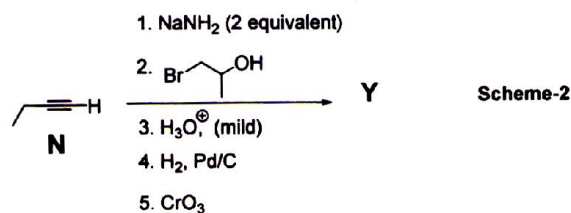
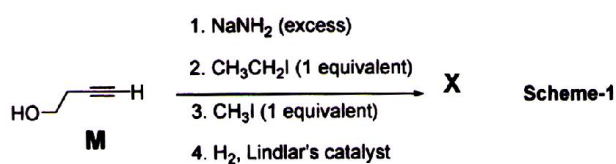
- (a) Statement I is true; Statement II is true; Statement II is a correct explanation of Statement I.
 (b) Statement I is true; Statement II is true; Statement II is not the correct explanation of Statement I.
 (c) Statement I is true; Statement II is false.
 (d) Statement I is false; Statement II is true.
38. **Statement I:** Addition of Br_2 to 1-butene gives two optical isomers.
Statement II: The product contains one asymmetric carbon. (1998)
39. **Statement I:** 1-butene on reaction with HBr in the presence of a peroxide produces 1-bromobutane.
Statement II: It involves the formation of a primary radical. (2000)
40. **Statement I:** Dimethyl sulphide is commonly used for the reduction of an alkene to get the carbonyl compound.
Statement II: It reduces the ozonide giving water soluble dimethyl sulphoxide and excess of it evaporates. (2001)
41. **Statement I:** Addition of bromine to trans-2 butene yields meso-2, 3-dibromo butane.
Statement II: Bromine addition to an alkene is an electrophilic addition. (2001)

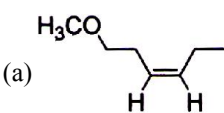
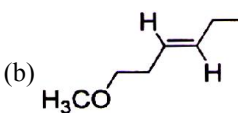
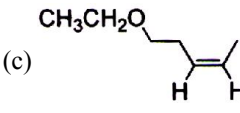
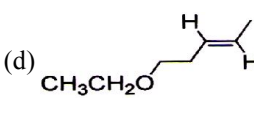
Comprehension Based Questions

An acyclic hydrocarbon P, having molecular formula C_6H_{10} , gave acetone as the only organic product through the following sequence of reactions, in which Q is an intermediate organic compound.



42. The structure of compound P is
 (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-\text{C}\equiv\text{C}-\text{H}$
 (b) $\text{H}_3\text{CH}_2\text{C}-\text{C}\equiv\text{C}-\text{CH}_2\text{CH}_3$
 (c) 
 (d) 
43. The structure of the compound Q is
 (a) 
 (b) 
 (c) 
 (d) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$
- Schemes 1 and 2 describe sequential transformation of alkynes M and N. Consider only the **major products** formed in each step for both the schemes. (2014)



44. The product X is
 (a) 
 (b) 
 (c) 
 (d) 
45. The correct statement with respect to product Y is
 (a) It gives a positive Tollens test and is a functional isomer of X.
 (b) It gives a positive Tollens test and is a geometrical isomer of X.
 (c) It gives a positive iodoform test and is a functional isomer of X.
 (d) It gives a positive iodoform test and is a geometrical isomer of X.
46. Hydrogen bonding plays a central role in the following phenomena (2016)
 (a) ice floats in water
 (b) higher Lewis basicity of primary amines than tertiary amines in aqueous solutions
 (c) formic acid is more acidic than acetic acid
 (d) dimerisation of acetic acid in benzene

Fill in the Blanks

47. is most acidic. (Ethane, Ethene, Ethyne) (1981)
48. Acetylene is treated with excess sodium in liquid ammonia. The product is reacted with excess methyl iodide. The final product is (1983)
49. The terminal carbon atom in 2-butene is hybridised. (1985)
50. Kolbe's electrolysis of potassium succinate gives CO_2 and (1993)
51. Addition of water to acetylene compounds is catalyzed by and (1993)
52. 1, 3-butadiene with bromine in molar ratio of 1 : 1 generate predominantly (1997)

True/False

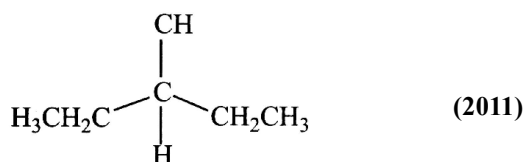
53. Moist ethylene can be dried by passing it through concentrated sulphuric acid. (1982)

Integer Type Questions

54. The total number of cyclic isomers possible for a hydrocarbon with the molecular formula C_4H_6 is (2010)
55. The maximum number of isomers (including stereoisomers) that are possible on mono-chlorination of the following compound, is

Subjective Questions

56. Give one characteristic test which would distinguish C_4H_4 from C_2H_2 . (1979)
57. Outline the reaction sequence for the conversion of ethene to ethyne (the number of steps should not be more than two). (1981)
58. Write the structural formula of the major product in each of the following cases
- Ethene mixed with air is passed under pressure over a silver catalyst.
 - The compound obtained by hydration of ethyne is treated with dilute alkali. (1981)
59. One mole of a hydrocarbon 'A' reacts with one mole of bromine giving a dibromo compound, $C_5H_{10}Br_2$. Compound 'A' on treatment with cold dilute alkaline potassium



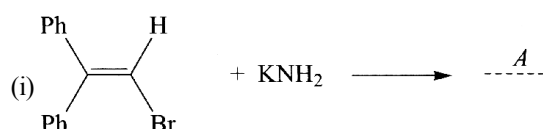
- permanganate solution forms a compound, $C_5H_{12}O_2$. On ozonolysis 'A' gives equimolar quantities of propanone and ethanal. Deduce the structural formula of 'A'. (1981)
60. State with balanced equation, what happens when "propene is bubbled through a hot aqueous solution of potassium permanganate."? (1982)
61. Give reasons for the following in one or two sentences:
- Methane does not react with chlorine in the dark.
 - Propene reacts with HBr to give isopropyl bromide but does not give n-propyl bromide. (1983)
62. Following statements are true, only under some specific conditions. Write the condition for each sub question in not more than two sentences:
- 2-methyl propene can be converted into isobutyl bromide by hydrogen bromide.
 - Ethyne and its derivatives will give white precipitate with ammoniacal silver nitrate solution. (1984)
63. Give the chemical test to distinguish between 2-butyne and 1-butyne. (1985)
64. How would you convert acetylene to acetone? (1985)
65. A white precipitate was formed slowly when silver nitrate was added to compound A with molecular formula $C_6H_{13}Cl$.

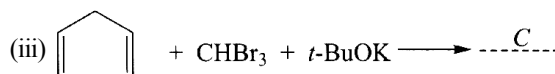
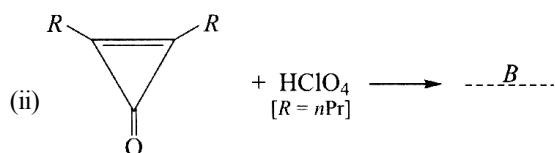
Compound A on treatment with hot alcoholic potassium hydroxide gave a mixture of two isomeric alkenes B and C, having formula C_6H_{12} . The mixture of B and C, on ozonolysis, furnished four compounds

- CH_3CHO
- C_2H_5CHO
- CH_3COCH_3
- $CH_3 - \underset{\text{CH}_3}{\text{CH}} - CHO$

What are the structures of A, B and C? (1986)

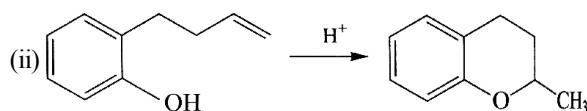
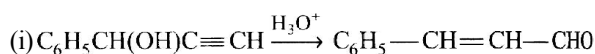
66. Give a chemical test and the reagents used to distinguish between cyclohexane and cyclohexene. (1991)
67. Write the balanced chemical equation for the following "Ethylene glycol is obtained by the reaction of ethylene with potassium permanganate." (1991)
68. Give the structures of A, B and C (explanation are not required)
- A (C_4H_8) which adds on HBr in the presence and in the absence of peroxide to give same product.
 - B (C_4H_8) which when treated with H_2O/H_2SO_4 gives $C_4H_{10}O$ which cannot be resolved into optical isomers.
 - C (C_6H_{12}), an optically active hydrocarbon which on catalytic hydrogenation gives an optically inactive compound C_6H_{14} . (1993)
69. When gas A is passed through dry KOH at low temperature, a deep red coloured compound B and a gas C are obtained. The gas A, on reaction with but-2-ene, followed by treatment with Zn/H_2O yields acetaldehyde. Identify A, B and C. (1994)
70. An organic compound E (C_5H_8) on hydrogenation gives compound F (C_5H_{12}). Compound E on ozonolysis gives formaldehyde and 2-ketopropanal. Deduce the structure of compound E. (1995)
71. Write down the structure of the stereoisomers formed when cis-2-butene is reacted with bromine. (1995)
72. Give the structure of the major organic products obtained from 3-ethyl-2-pentene under each of the following reaction conditions: (1996)
- HBr in the presence of peroxide
 - Br_2/H_2O
 - $Hg(OAc)_2/H_2O, NaBH_4$
73. An alkyl halide, X, of formula $C_6H_{13}Cl$ on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z (C_6H_{12}). Both alkenes on hydrogenation gives 2, 3-dimethyl butane. Predict the structures of X, Y and Z. (1996)
74. Complete the following, giving the structures of the principal organic products: (1997)



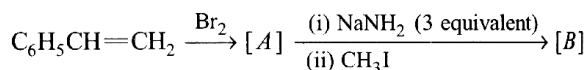


75. The hydrocarbon A, adds one mole of hydrogen in the presence of a platinum catalyst to form n-hexane. When A is oxidised vigorously with KMnO_4 , a single carboxylic acid, containing three carbon atoms, is isolated. Give the structure of A and explain. (1997)

76. Write the intermediate steps for each of the following reactions:

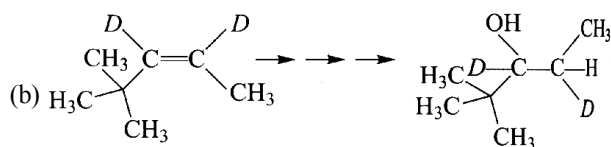
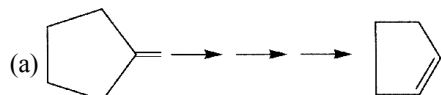


77. Complete the following reactions with appropriate structures of products/reagents:

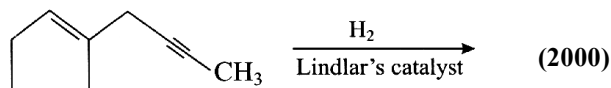


(1998)

78. Complete the following reactions with appropriate reagents:

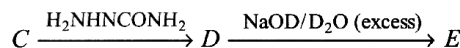
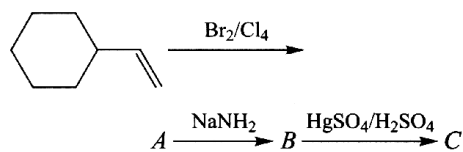


79. What would be the major product in the following reaction?

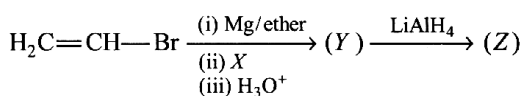


80. An alkene (A) $\text{C}_{16}\text{H}_{16}$ on ozonolysis gives only one product (B) $\text{C}_8\text{H}_8\text{O}$. Compound (B) on reaction with NaOH/I_2 yields sodium benzoate. Compound (B) reacts with $\text{KOH}/\text{NH}_2\text{NH}_2$ yielding a hydrocarbon (C) C_8H_{10} . Write the structures of compounds (B) and (C). Based on this information two isomeric structures and identify the isomer which on catalytic hydrogenation ($\text{H}_2/\text{Pd-C}$) gives a racemic mixture. (2001)

81. (a) Identify (A), (B), (C), (D) and (E) in the following schemes and write their structures:

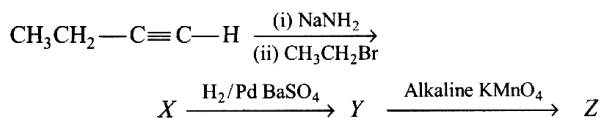


- (b) Identify (X), (Y) and (Z) in the following synthetic scheme and write their structures. Explain the formation of labelled formaldehyde ($\text{H}_2\text{C}^*\text{O}$) as one of the products when compound (Z) is treated with HBr and subsequently ozonolysed. Mark the C^* carbon in the entire scheme. $\text{BaC}^*\text{O}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{X}) \text{ gas } [\text{C}^* \text{ denotes } \text{C}^{14}]$



(2001)

82. Identify X, Y and Z in the following synthetic scheme and write their structures.



(2002)

Is the compound Z optically active? Justify your answer

83. A biologically active compound, Bombykol ($\text{C}_{16}\text{H}_{30}\text{O}$) is obtained from a natural source. The structure of the compound is determined by the following reactions.

- (a) On hydrogenation, Bombykol gives a compound A, $\text{C}_{16}\text{H}_{34}\text{O}$, which reacts with acetic anhydride to give an ester.
(b) Bombykol also reacts with acetic anhydride to give another ester, which on oxidative ozonolysis ($\text{O}_3/\text{H}_2\text{O}_2$) gives a mixture of butanoic acid, oxalic acid and 10-acetoxy decanoic acid.

Determine the number of double bonds in Bombykol. Write the structures of compound A and Bombykol. How many geometrical isomers are possible for Bombykol? (2002)

84. Monomer A of a polymer on ozonolysis yields two moles of HCHO and one mole of CH_3COCHO .

- (a) Deduce the structure of A.
(b) Write the structure of all "cis" form of polymer of compound A. (2005)

Answer Key

Exercise - 1 : (Basic Objective Questions)

1. (a)	2. (c)	3. (b)	4. (b)	5. (b)	6. (c)	7. (a)	8. (a)	9. (c)	10. (a)
11. (a)	12. (b)	13. (d)	14. (c)	15. (b)	16. (b)	17. (c)	18. (c)	19. (d)	20. (b)
21. (c)	22. (d)	23. (c)	24. (a)	25. (b)	26. (c)	27. (b)	28. (a)	29. (a)	30. (b)
31. (b)	32. (d)	33. (b)	34. (c)	35. (b)	36. (b)	37. (c)	38. (d)	39. (d)	40. (a)
41. (c)	42. (b)	43. (c)	44. (c)	45. (b)	46. (b)	47. (c)	48. (c)	49. (c)	50. (c)
51. (c)	52. (a)	53. (d)	54. (b)	55. (a)	56. (c)	57. (b)	58. (b)	59. (a)	60. (b)
61. (b)	62. (a)	63. (c)	64. (a)	65. (b)	66. (a)	67. (a)	68. (d)	69. (b)	70. (d)
71. (a)	72. (b)	73. (a)	74. (a)	75. (b)	76. (c)	77. (c)	78. (a)	79. (c)	80. (a)
81. (b)	82. (d)	83. (a)	84. (c)	85. (b)	86. (a)	87. (b)	88. (a)	89. (b)	90. (d)
91. (b)	92. (c)	93. (c)	94. (b)	95. (b)	96. (d)	97. (c)	98. (d)	99. (c)	100. (c)

Exercise - 2 : (Previous Year JEE Mains Questions)

1. (c)	2. (a)	3. (d)	4. (d)	5. (d)	6. (a)	7. (d)	8. (d)	9. (b)	10. (c)
11. (a)	12. (d)	13. (d)	14. (c)	15. (c)	16. (c)	17. (c)	18. (b)	19. (d)	20. (a)
21. (b)	22. (d)	23. (d)	24. (c)	25. (d)	26. (a)	27. (d)	28. (d)	29. (b)	30. (d)
31. (c)	32. (b)	33. (b)	34. (c)	35. (a)	36. (d)				

JEE Mains Online

1. (d)	2. (d)	3. (c)	4. (d)	5. (c)	6. (d)	7. (a)
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Exercise - 3 : (Advanced Objective Questions)

1. (a)	2. (d)	3. (a)	4. (b)	6. (b)	7. (a)	8. (d)	9. (a)	10. (d)	11. (c)
12. (0001)	13. (abd)	14. (d)	15. (c)	16. (d)	17. (a)	18. (b)	19. (b)		
20. (a – Q,R; b – Q,R; c – P; d – P, S) 21. (a – P, S; b – P, Q, R; C – Q, R; d – P, Q, S)									
22. (a – P, Q, R; b – P, Q, R, S; c – P, Q, R, S; d – Q)					23. (b)	24. (a)	25. (a)	26. (a)	27. (a)

28. (d) 29. (c) 30. (b) 31. (acd) 32. (a) 33. (d) 34. (a) 35. (abc) 36. (c) 37. (c)
38. (b) 39. (abc) 40. (0005) 41. (b) 42. (abd) 43. (b) 44. (d) 46. (c) 47. (ab) 48. (a)
49. (acd) 50. (abcd) 51. (ab) 52. (bc) 53. (b) 54. (d) 55. (a) 56. (b) 57. (c) 58. (b)
59. (a) 60. (b) 61. (acd) 62. (abc) 62. (c) 63. (c) 64. (c) 65. (0006) 66. (c) 67. (b)
68. (b) 69. (b) 70. (c) 71. (b) 72. (d) 73. (a) 74. (a, b, c, d)
75. (a – q, r ; b – p, r ; c – q, s ; d – p, r) 76. (a) 77. (d) 78. (c) 79. (d) 80. (2143)
81. (1324) 82. (b) 83. (a) 84. (cd) 85. (b) 86. (acd) 87. (ac) 88. (c) 89. (c) 90. (b)
91. (a) 92. (b) 93. (c) 94. (0005) 95. (0005) 96. (0006) 97. (a) 98. (0002) 99. (0004) 100. (b)

Exercise - 4 : (Previous Year JEE Advanced Questions)

1. (b) 2. (d) 3. (d) 4. (d) 5. (c) 6. (c) 7. (a) 8. (a) 9. (b) 10. (b)
11. (d) 12. (c) 13. (b) 14. (d) 15. (a) 16. (c) 17. (c) 18. (d) 19. (a) 20. (a)
21. (d) 22. (b) 23. (b) 24. (c) 25. (d) 26. (b) 27. (a) 28. (a) 29. (a) 30. (c)
31. (d) 32. (a) 33. (a) 34. (b) 35. (b) 36. (a) 37. (d) 38. (a) 39. (c) 40. (a)
41. (b) 42. (d) 43. (b) 44. (a) 45. (c) 46. (a,b,d) 47. (Terminal alkyne (ethyne))
48. (2-butyne) 49. (2-butene) 50. (ethene) 51. (H_2SO_4 , HgSO_4) 52. (3,4-dibromo-1-butene)
53. (F) 54. (0005) 55. (0008)

Dream on !!

